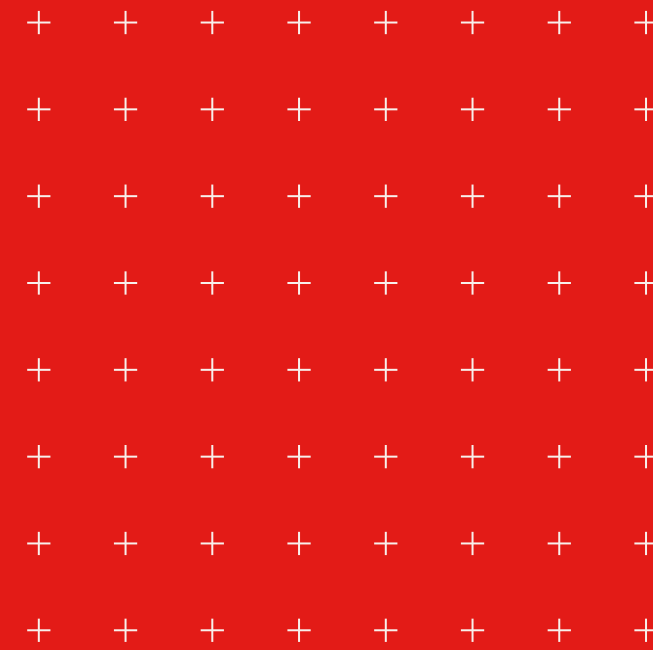




27/01/2026
Research and
innovation
project,
INSA Toulouse

Metamodeling for flooding alarm system by integrating functional inputs and outputs

Victor Davodeau, Paul Slisse

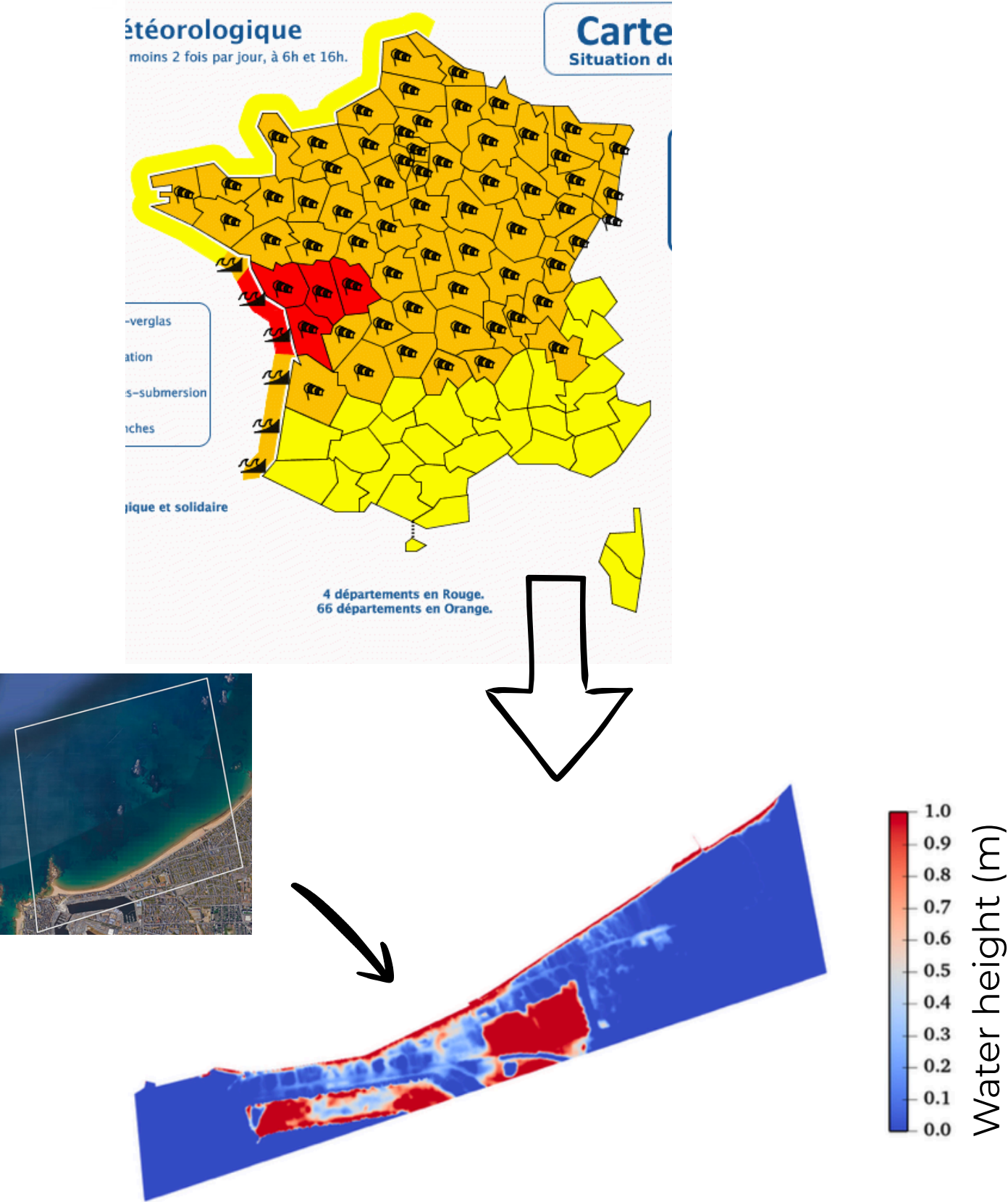
5th year Department of Applied Mathematics, INSA Toulouse, France

Tutors: Pascal Noble, Olivier Roustant, Rémy Baraille
Toulouse Institute of Mathematics - INSA Toulouse - SHOM, France



**Currently we can't predict the risk of coastal flooding
locally and in real time**

Develop a new model able to predict instantly the water level in a city



Real time prediction

Time to work !

Physical models are too slow (10 minutes)

Outline

I

Metamodelisation and gaussian process

What is metamodelisation?
Gaussian processes
How can this solve our problem

II

Methodology and toy model

From a scalar input to a scalar output
Adding temporality to the input
Adding temporality to the output

III

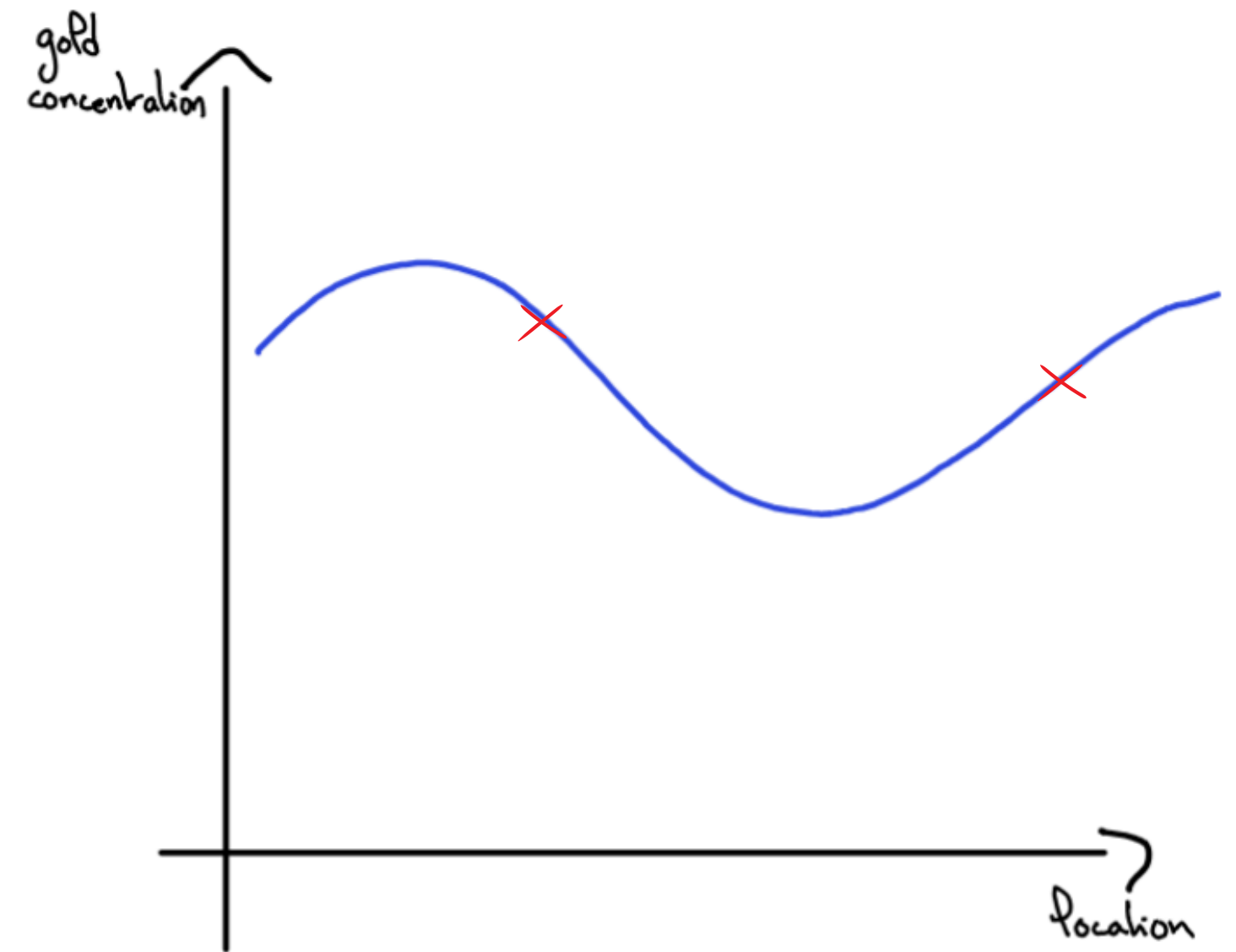
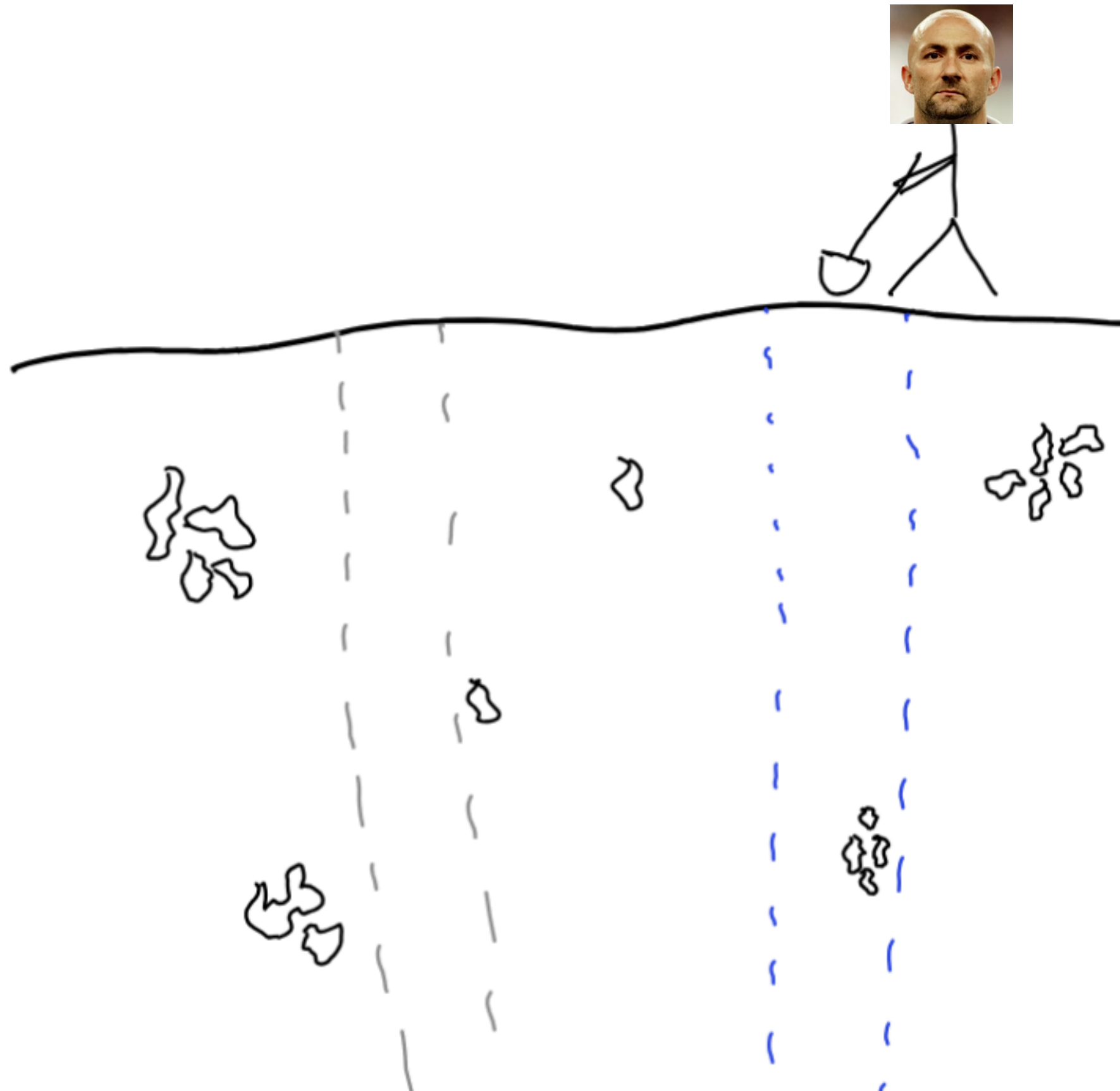
Saint Malo

Application on a real case :
The bay of Saint-Malo

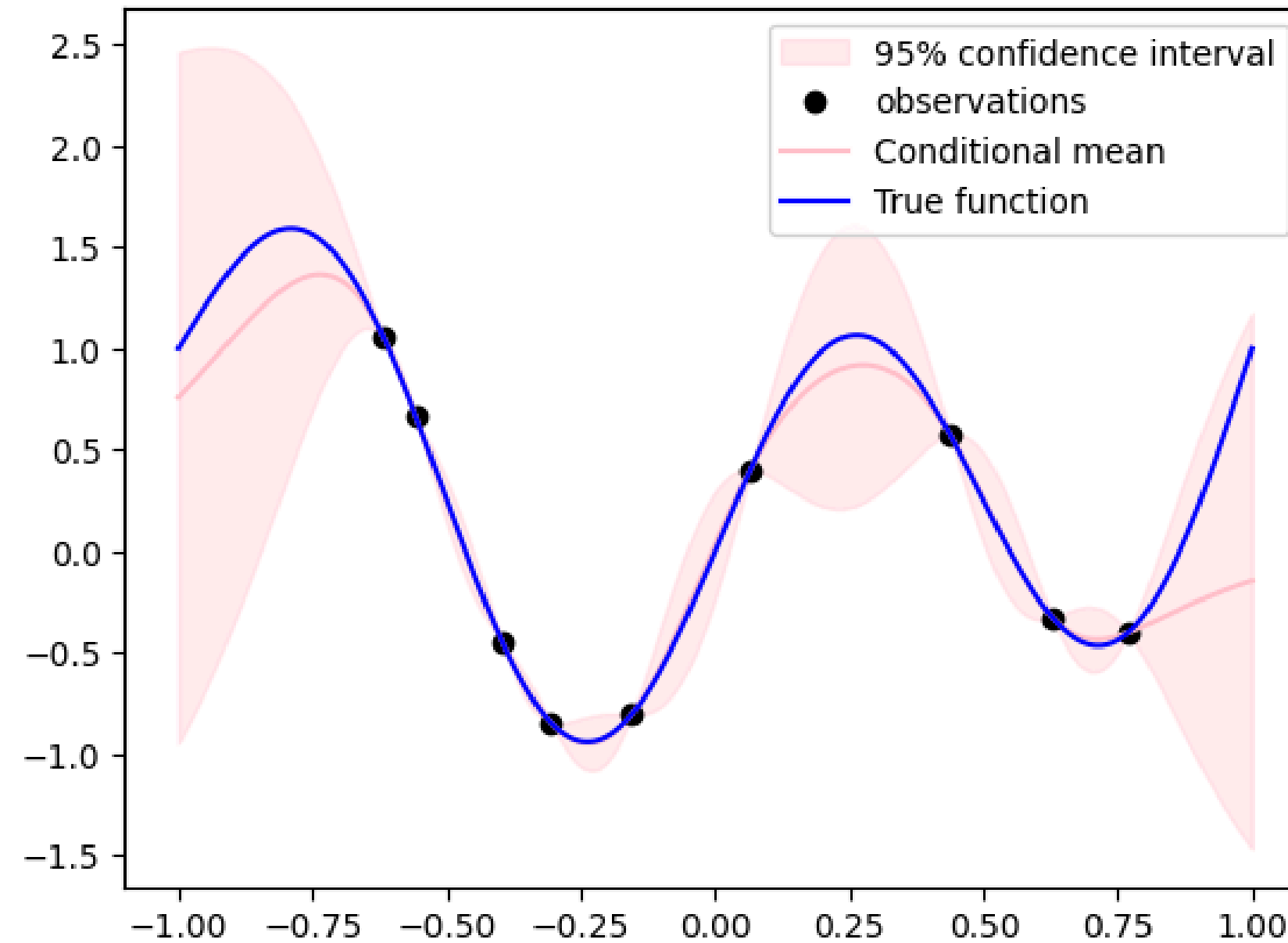
Metamodelisation and gaussian process

- Grasping the idea with a simple example
- Quick mathematical definition
- Link with our problem

Fabien the gold miner

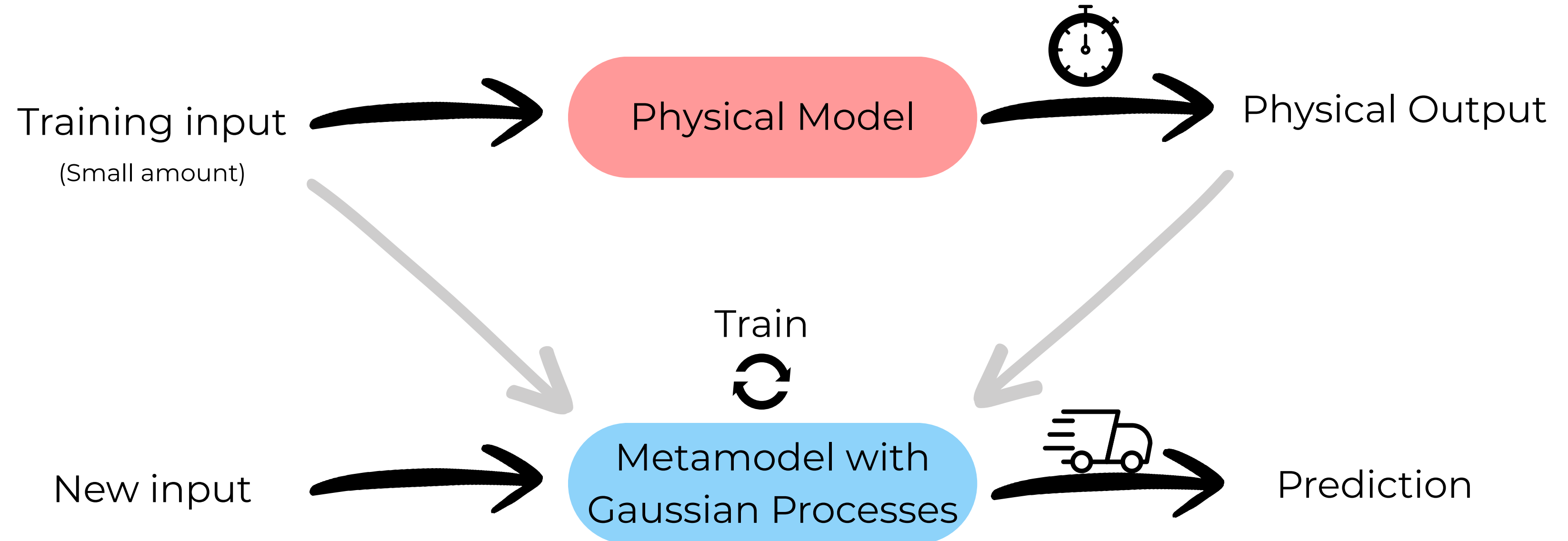


Few observations leads to good approximation !



$$\begin{aligned} m_c(x) &= m(x) + k(x, X)k(X, X)^{-1}(y - m(X)) \\ k_c(x, x') &= k(x, x') - k(x, X)k(X, X)^{-1}k(X, x') \end{aligned}$$

We can use metamodeling for our problem



Methodology and toy model

- Toy function
- How to simulate our data
- Reduce dimension of the output
- Adding temporality to the input
- Adding temporality to the output

Campbell, the **toy function** to test our models

I

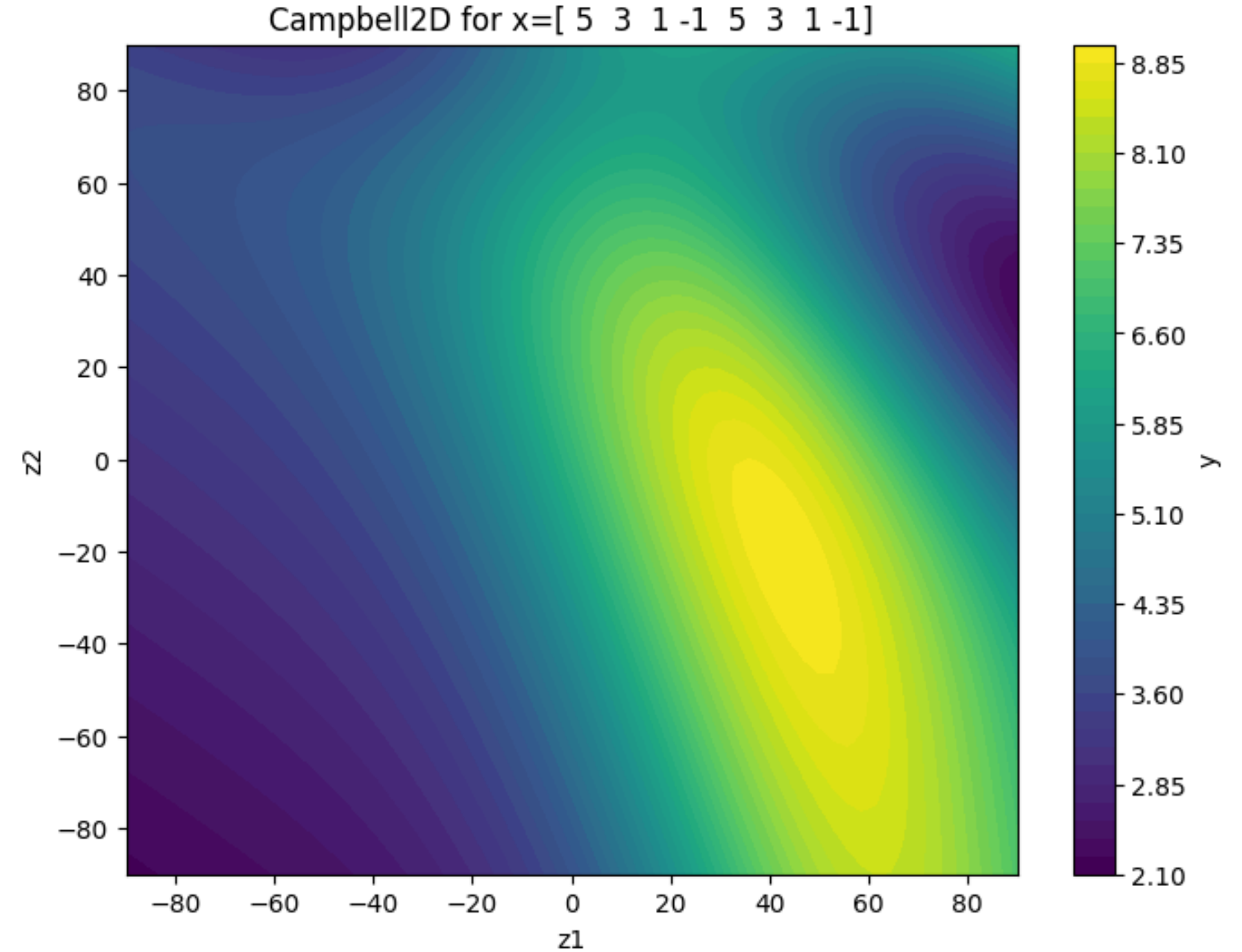
Methodology and toy model

III

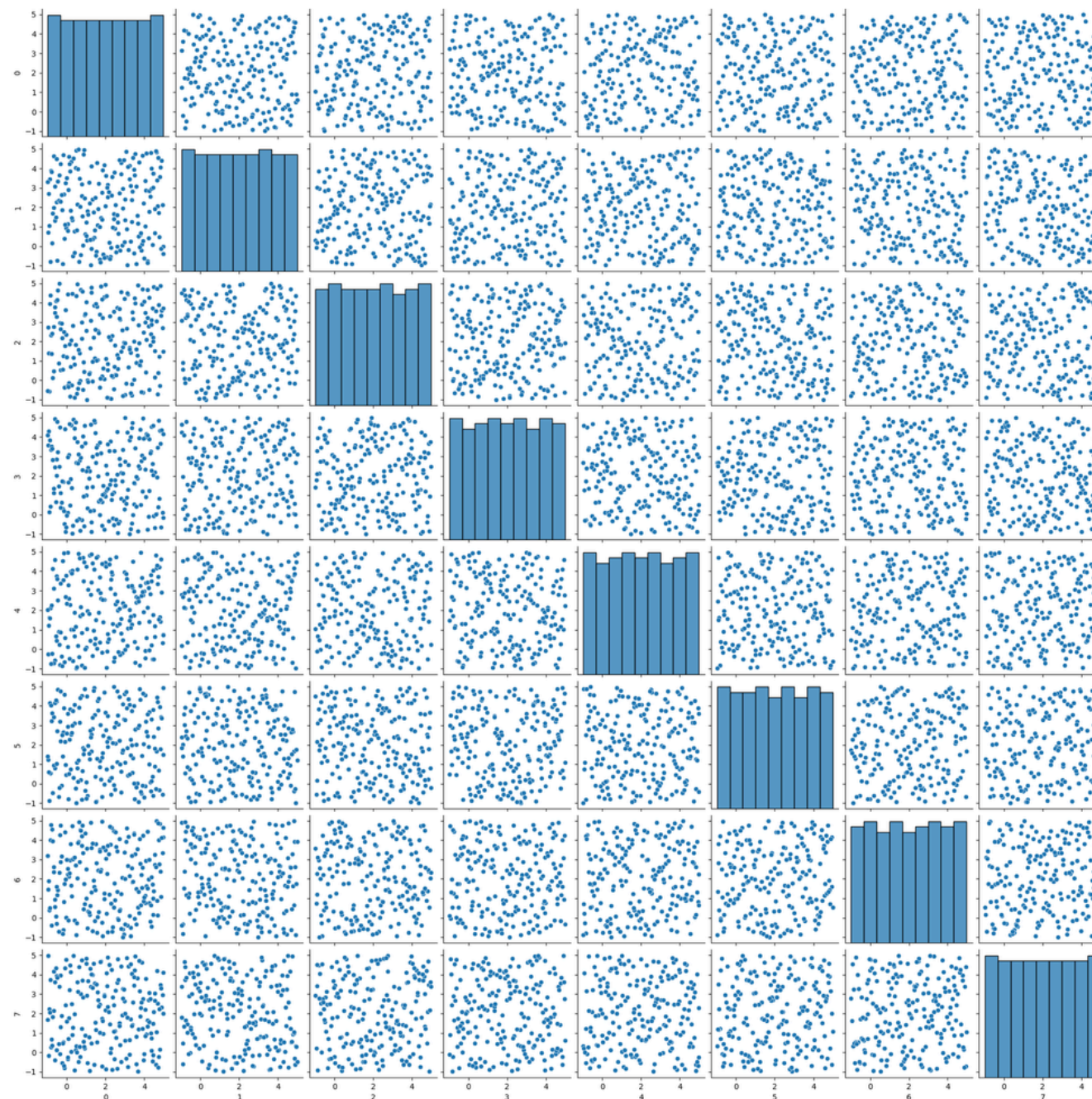
$$f : [-1, 5]^8 \longrightarrow L^2([-90, 90]^2)$$

$$x = (x_1, \dots, x_8) \longmapsto y_x(z)$$

$$\begin{aligned} y_x(z_1, z_2) = & x_1 \exp\left(-\frac{(0.8z_1 + 0.2z_2 - 10x_2)^2}{60x_1^2}\right) \\ & + (x_2 + x_4) \exp\left(\frac{(0.5z_1 + 0.5z_2)x_1}{500}\right) \\ & + x_5(x_3 - 2) \exp\left(-\frac{(0.4z_1 + 0.6z_2 - 20x_6)^2}{40x_5^2}\right) \\ & + (x_6 + x_8) \exp\left(\frac{(0.3z_1 + 0.7z_2)x_7}{250}\right). \end{aligned}$$



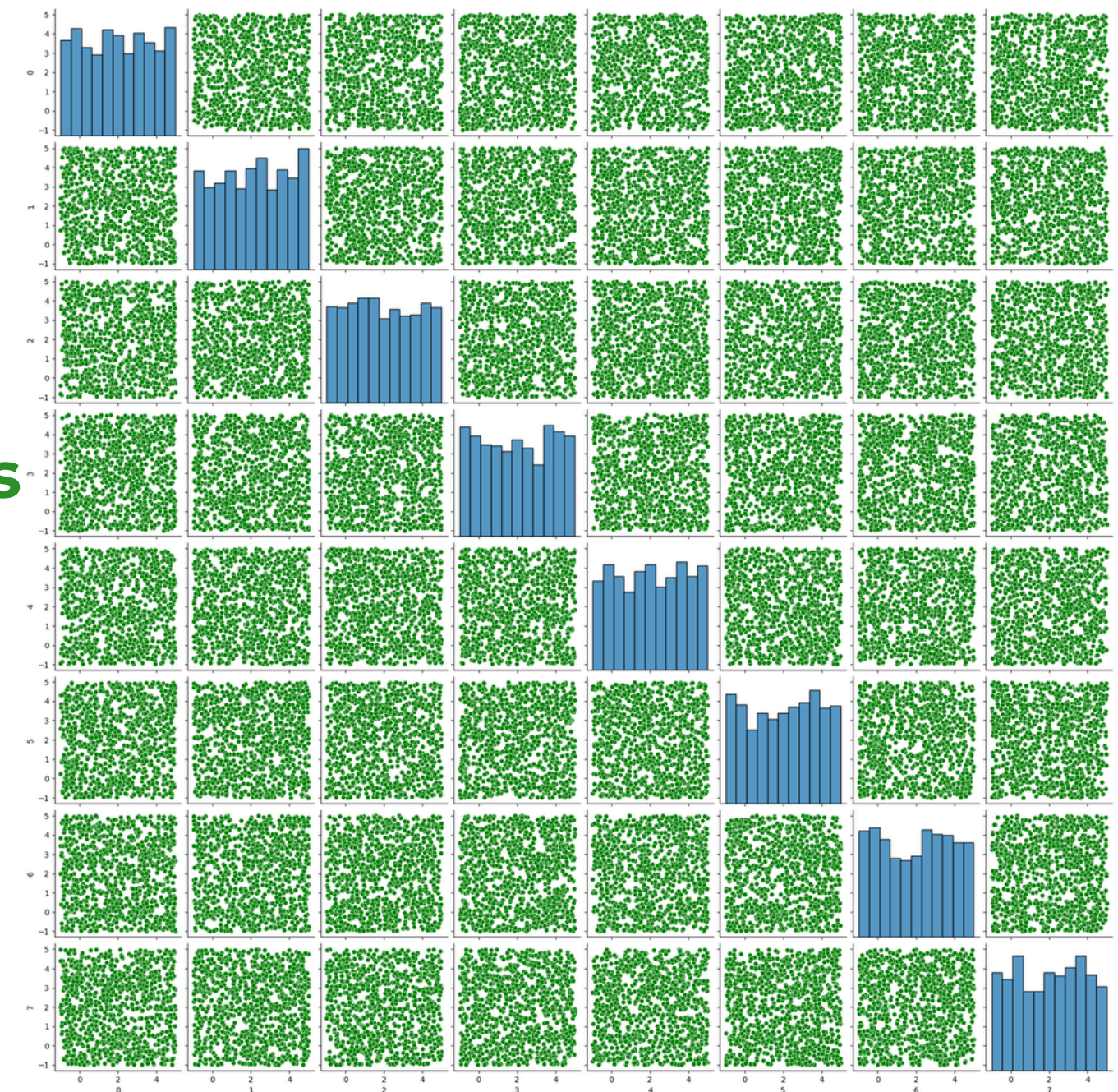
Latin Hypercube, the best way to **simulate our data**



Design of experiment of the 8 scalar input
for the training data

Train
200 points
Latin
Hypercube

Test
1000 points
Uniform
[-1,5]

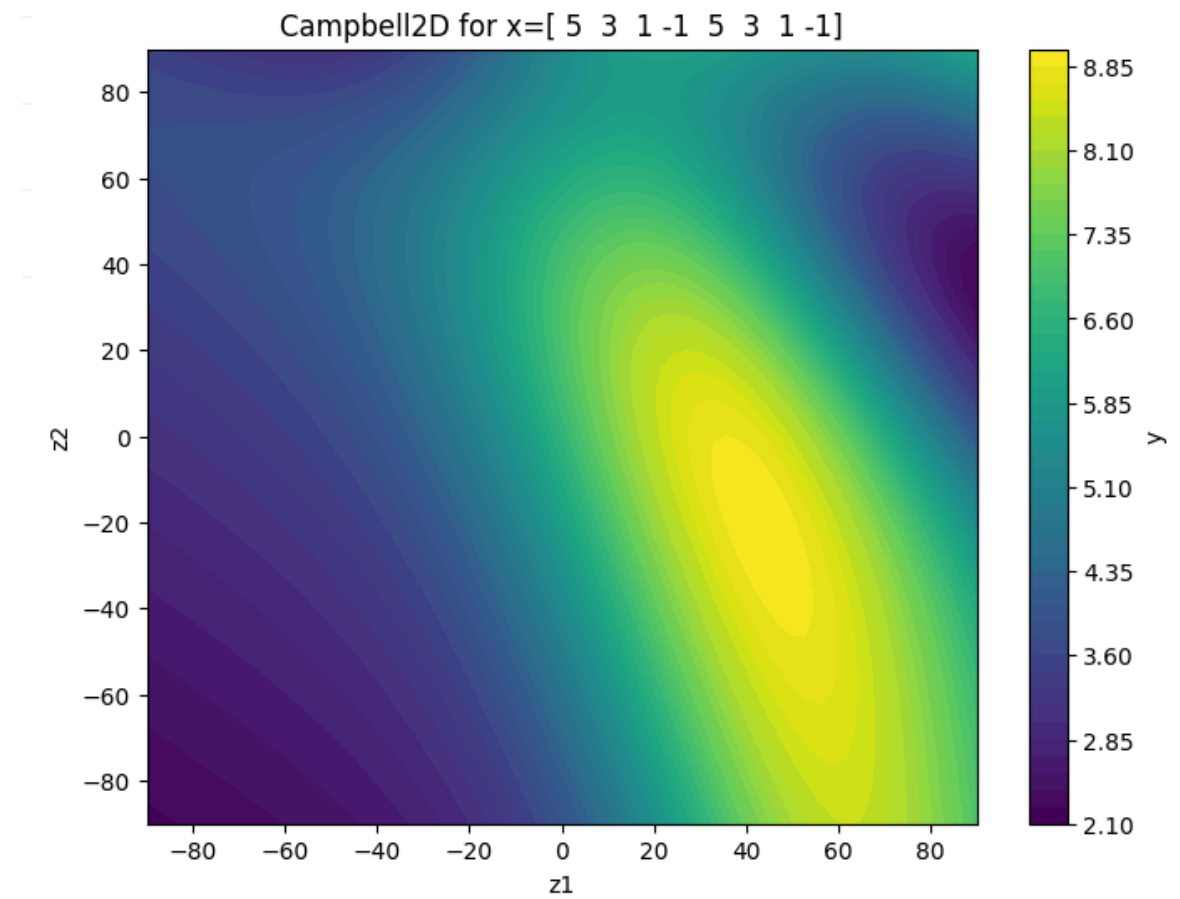


Design of experiment of the 8 scalar input
for the testing data

The dimensionality **problem**

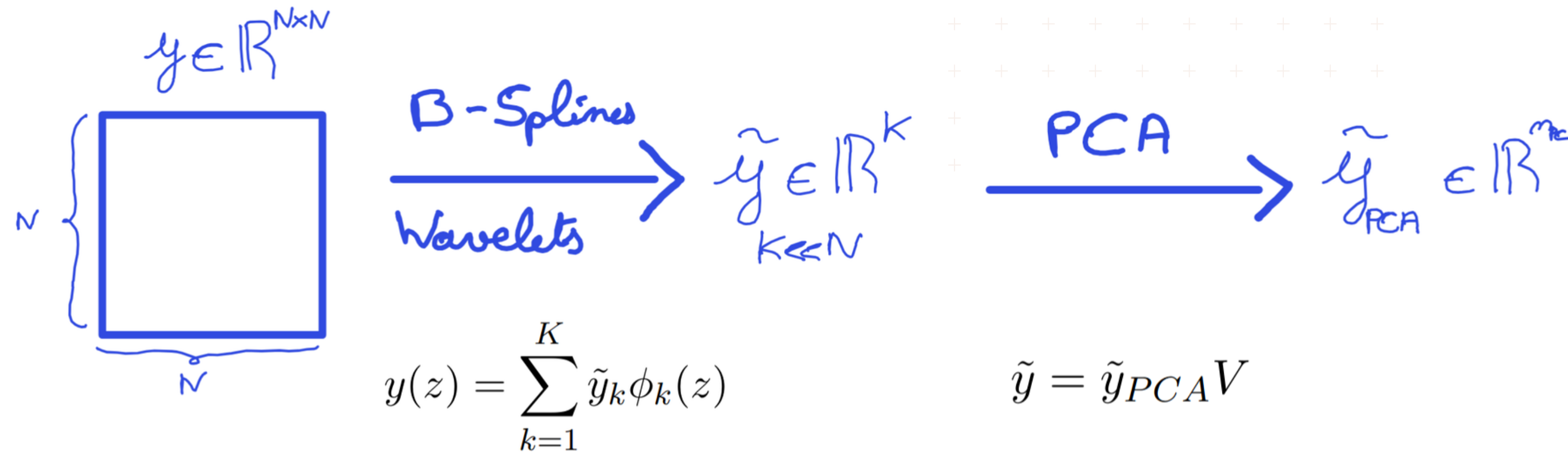
Scalar input
[5, 3, 1 -1, 5, 3, 1, -1]

Predict the output



Problem !
Dimension of the spatial output (the map) is
way too big !

3 ways to reduce the dimension : PCA, Wavelets and B-Splines

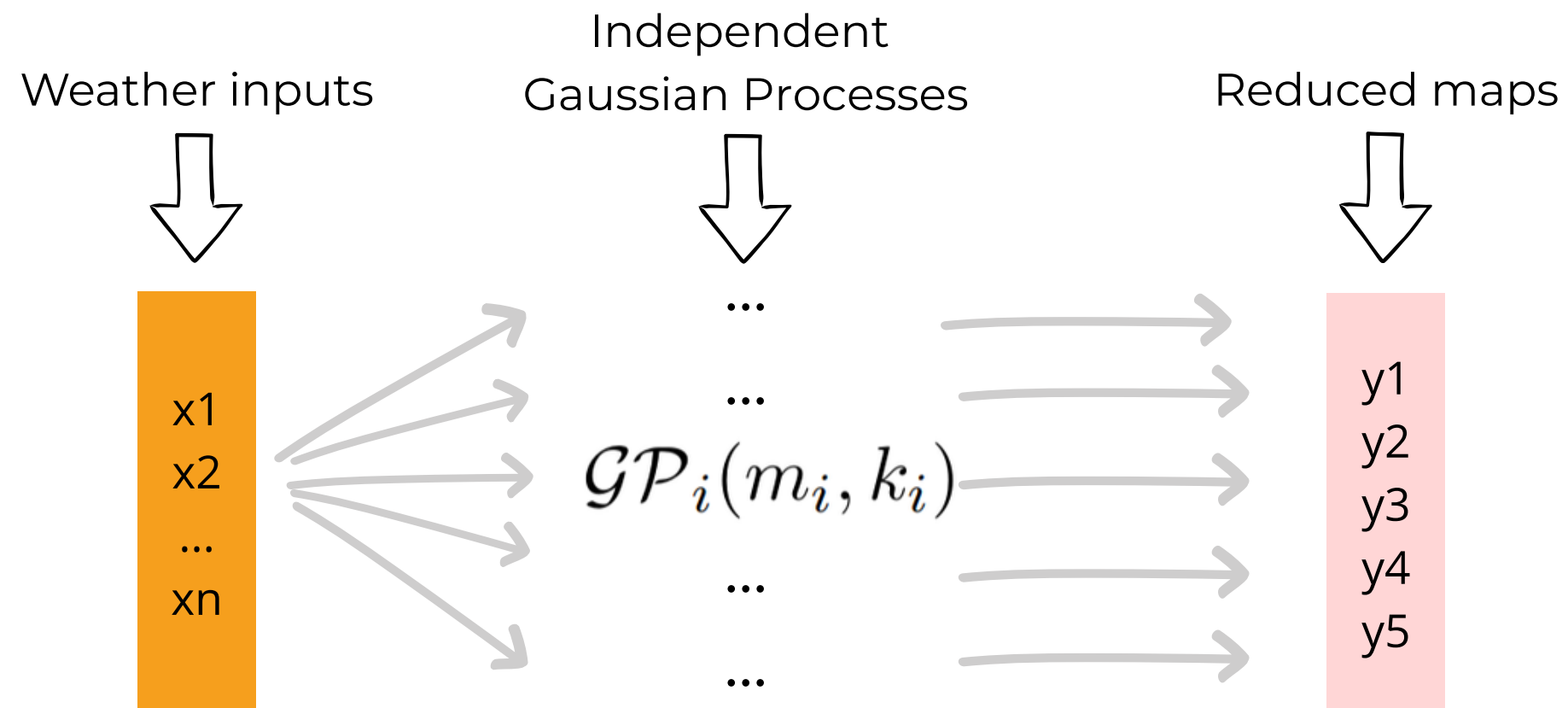


Start with a high
dimensional spatial
input

Conserve most information and
spatial dependences

End with a small 1D
Vector

The core of the metamodel : Gaussian Process regression



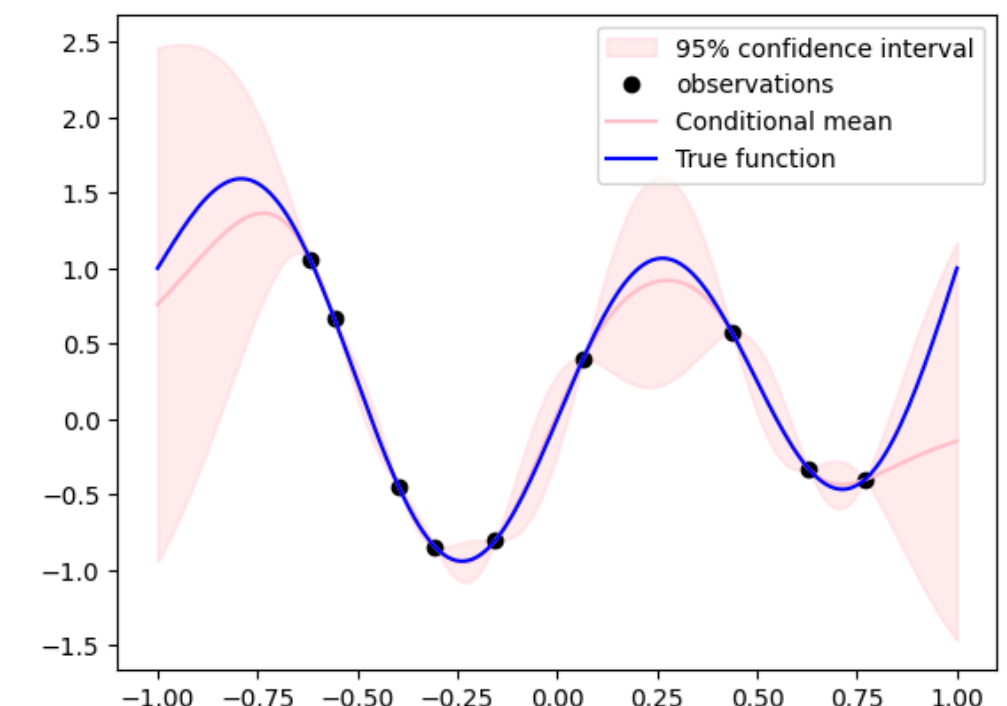
Squared exponential kernel

$$k_i(x, x') = \sigma_i^2 \exp \left(-\frac{\|x - x'\|_2^2}{2\ell_i^2} \right)$$

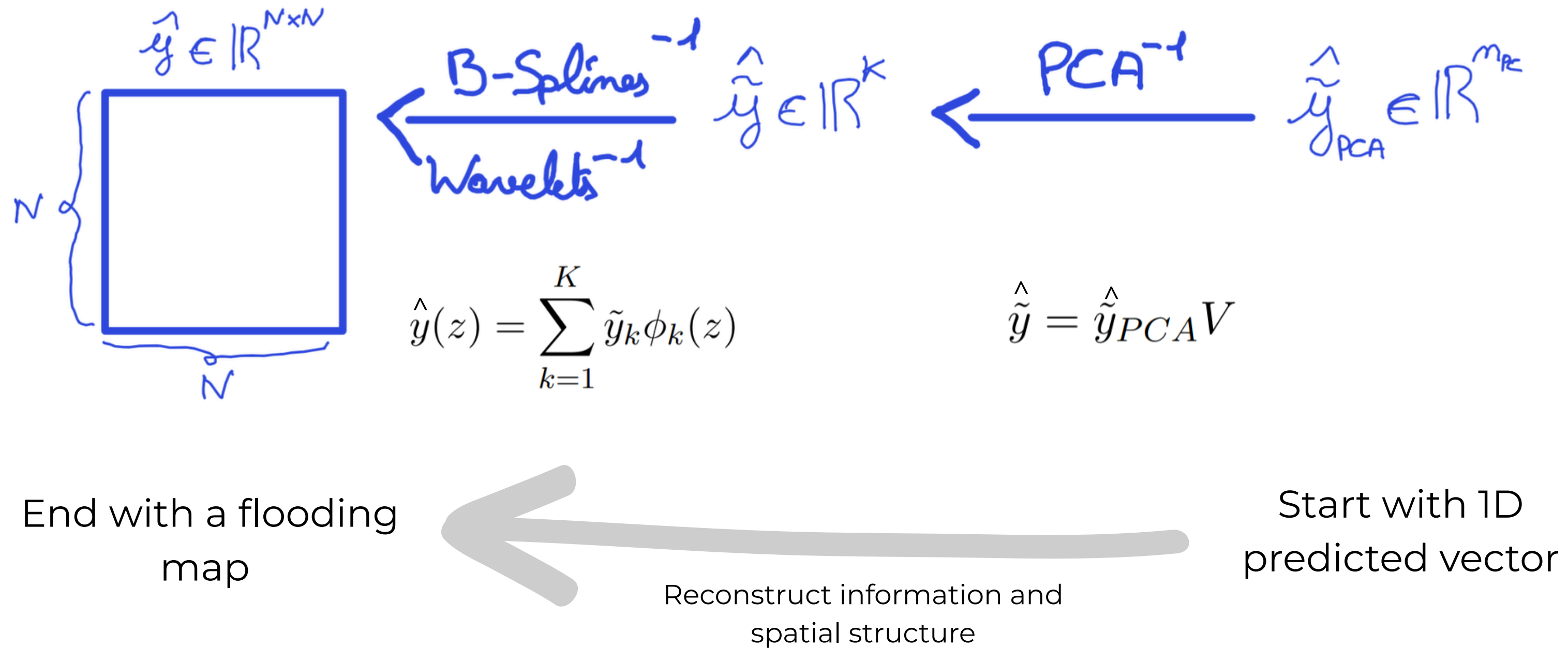
Tuning of hyperparameters with Maximum Likelihood Estimation

Inference: Gaussian sampling

$$\begin{aligned} m_c(x) &= m(x) + k(x, X)k(X, X)^{-1}(y - m(X)) \\ k_c(x, x') &= k(x, x') - k(x, X)k(X, X)^{-1}k(X, x') \end{aligned}$$



Apply the inverse pipeline for **inverse projection**



All 3 methods are **predicting** accurately

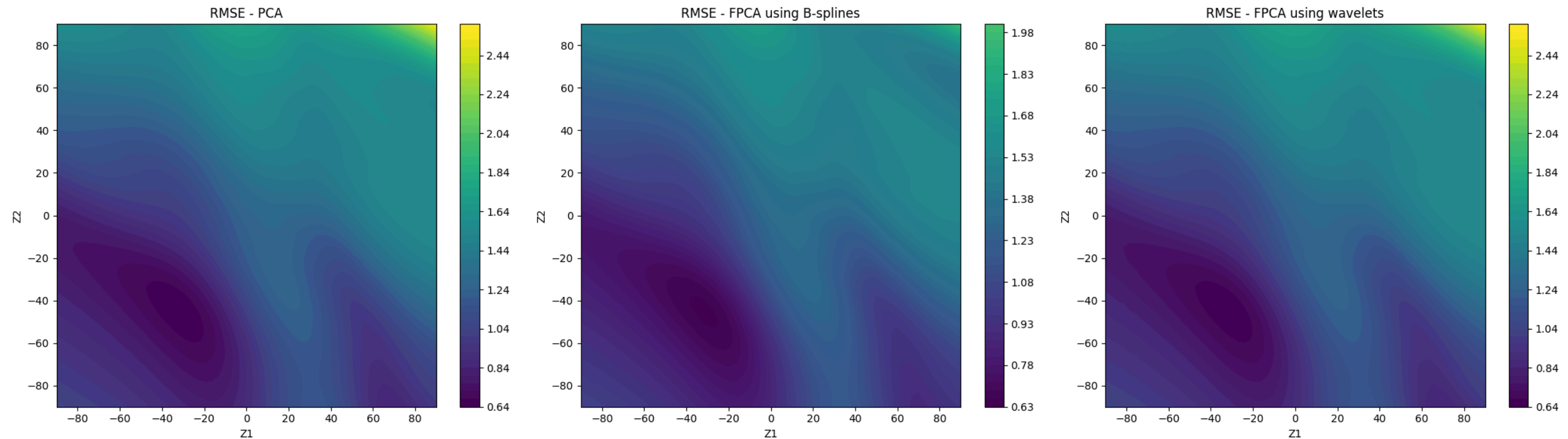
I

Methodology and toy model

III

- Generate train and test data ✓
- Use gaussian processes ✓
- Reduce output dimension ✓

$$\text{RMSE}_j = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{i,j} - \hat{y}_{i,j})^2}$$



Adding temporality to the input

I

Methodology and toy model

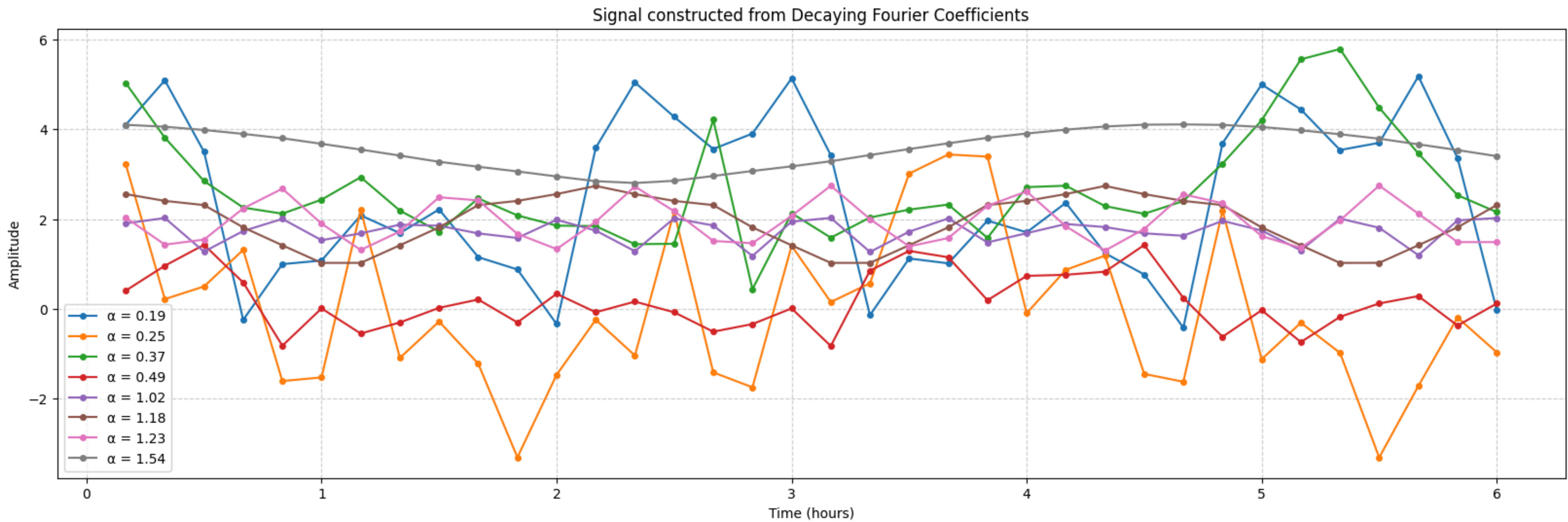
III

↪ $X = (f_1, f_2, \dots, f_8) \quad \mathbb{X} = (L^2([0, T]))^d$

each function f_i is defined on the time interval $[0, T]$

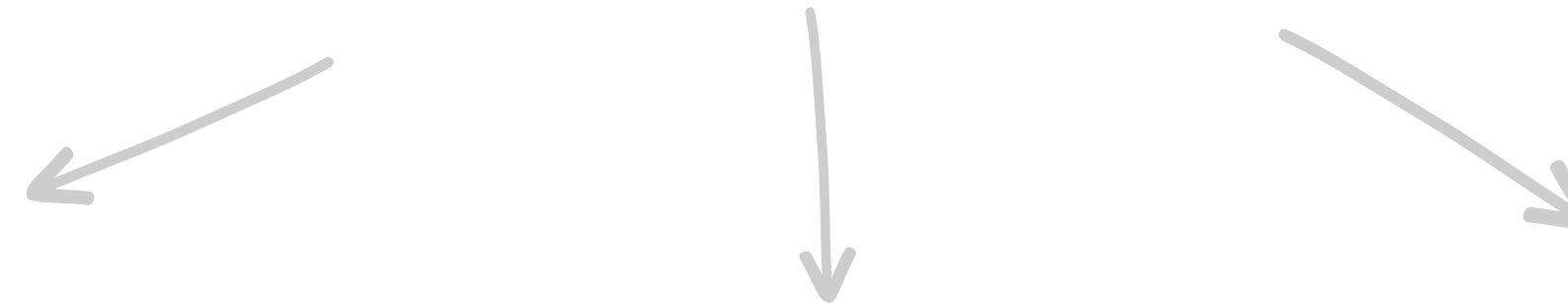
$$f_i(t) = 2 + A \sum_{k=0}^{M-1} \frac{a_{i,k}}{(1 + k^2)^{\alpha_i}} e^{i\omega_i k t}$$

We use Fourier coefficients to build f !

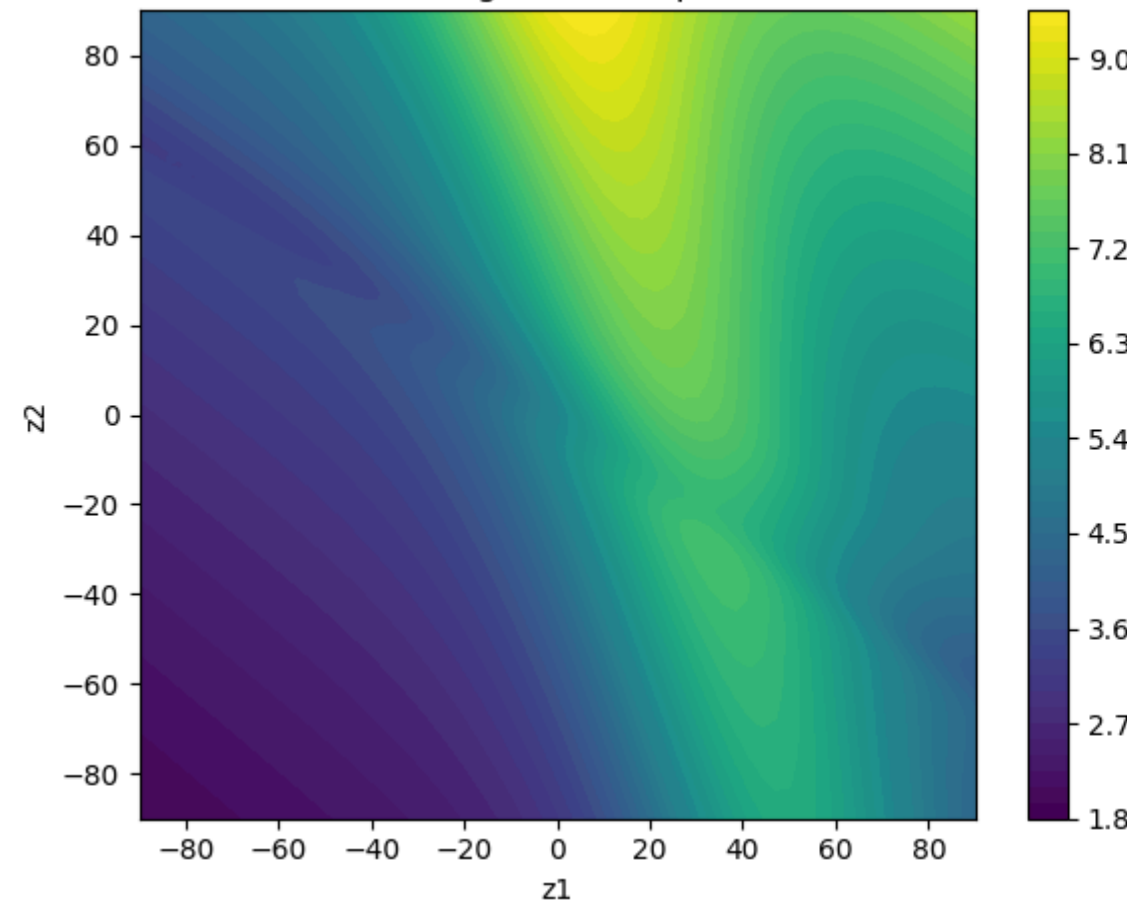


Adapt Campbell function

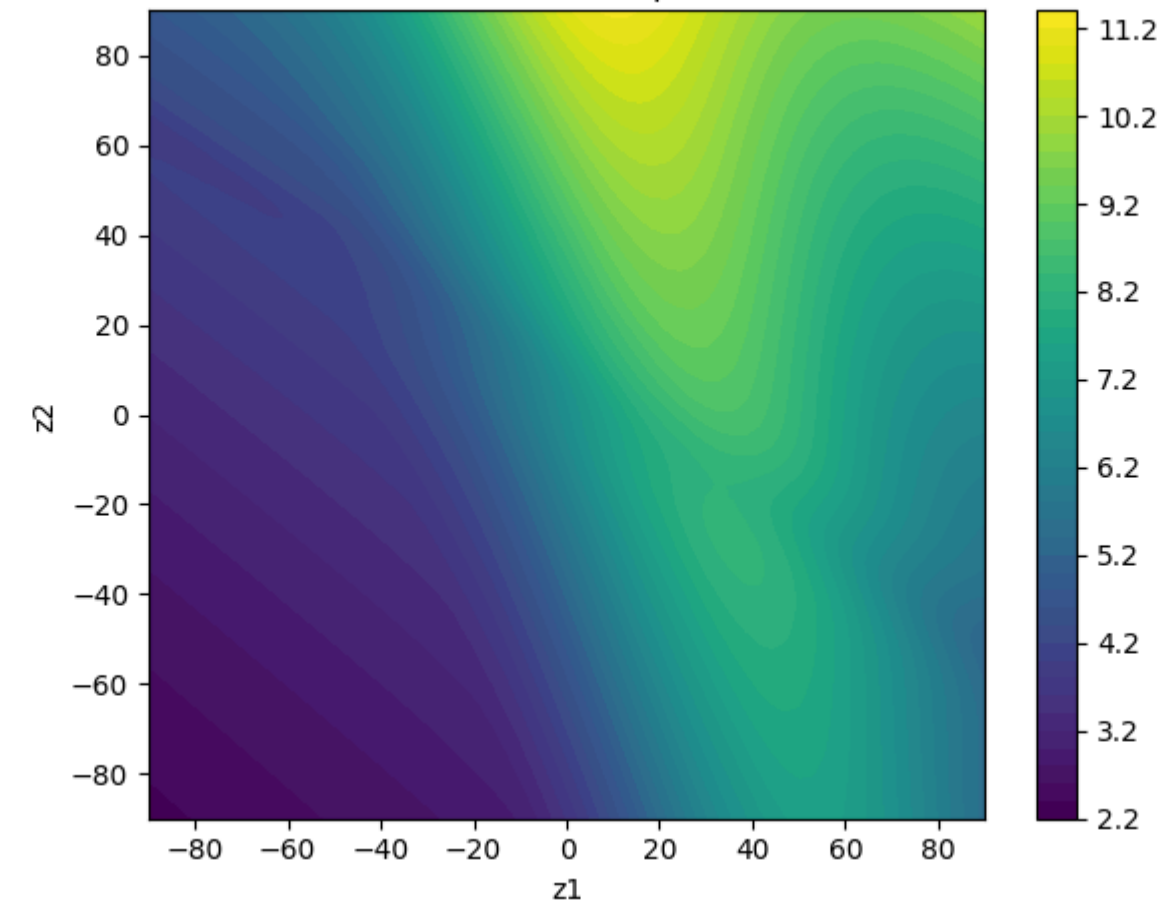
Compute Campbell at each time step
and retain summary statistics :
Average, max and min



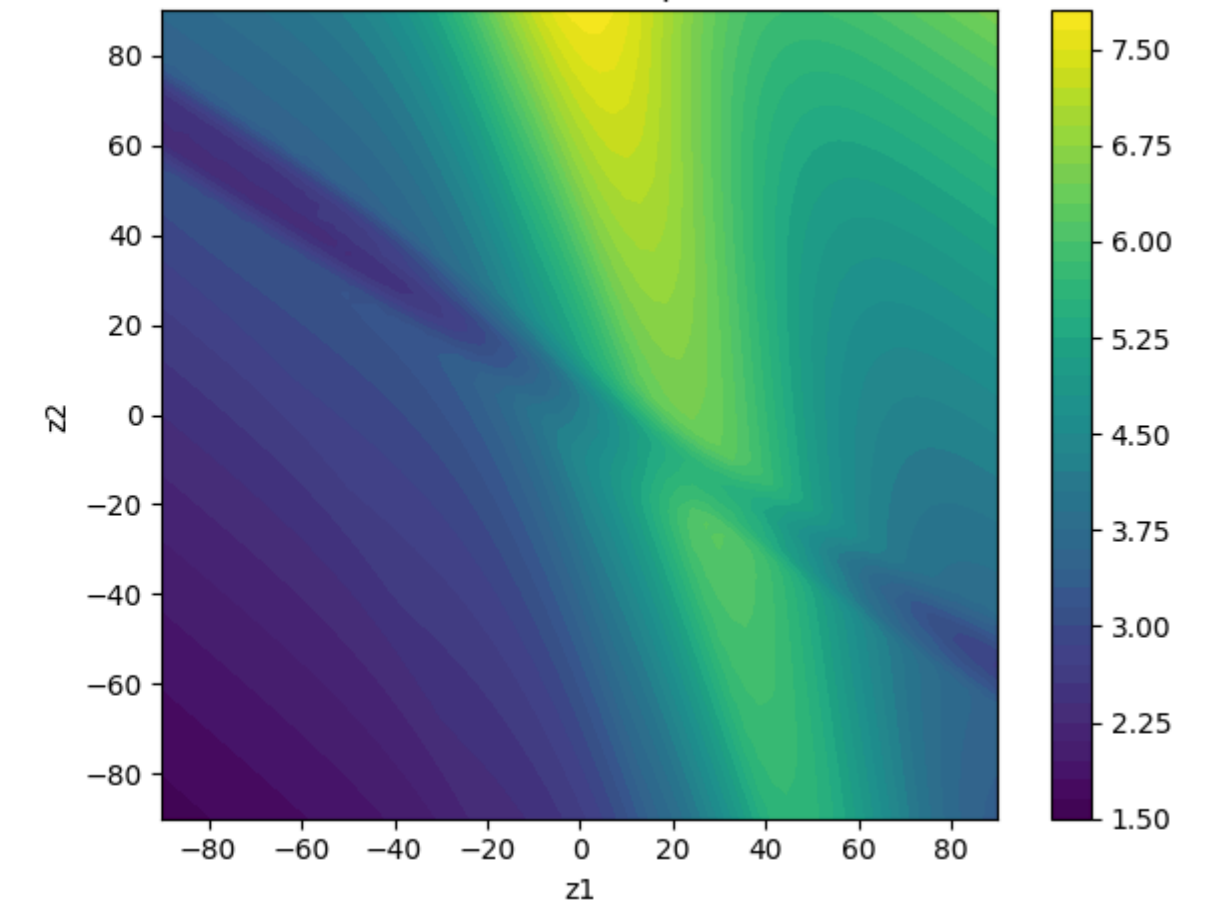
Average water depth



Max water depth



Min water depth



We can now generate outputs !

Adapt the kernel

$$X = (f_1, f_2, \dots, f_8) \quad \mathbb{X} = (L^2([0, T]))^d$$

Sobolev norm

temporal variations

$$\|X - X'\|_{H^1}^2 = \sum_{i=1}^d \|f_i - g_i\|_{L^2([0, T])}^2 + \alpha \sum_{i=1}^d \|f'_i - g'_i\|_{L^2([0, T])}^2$$

discrepancies in amplitude

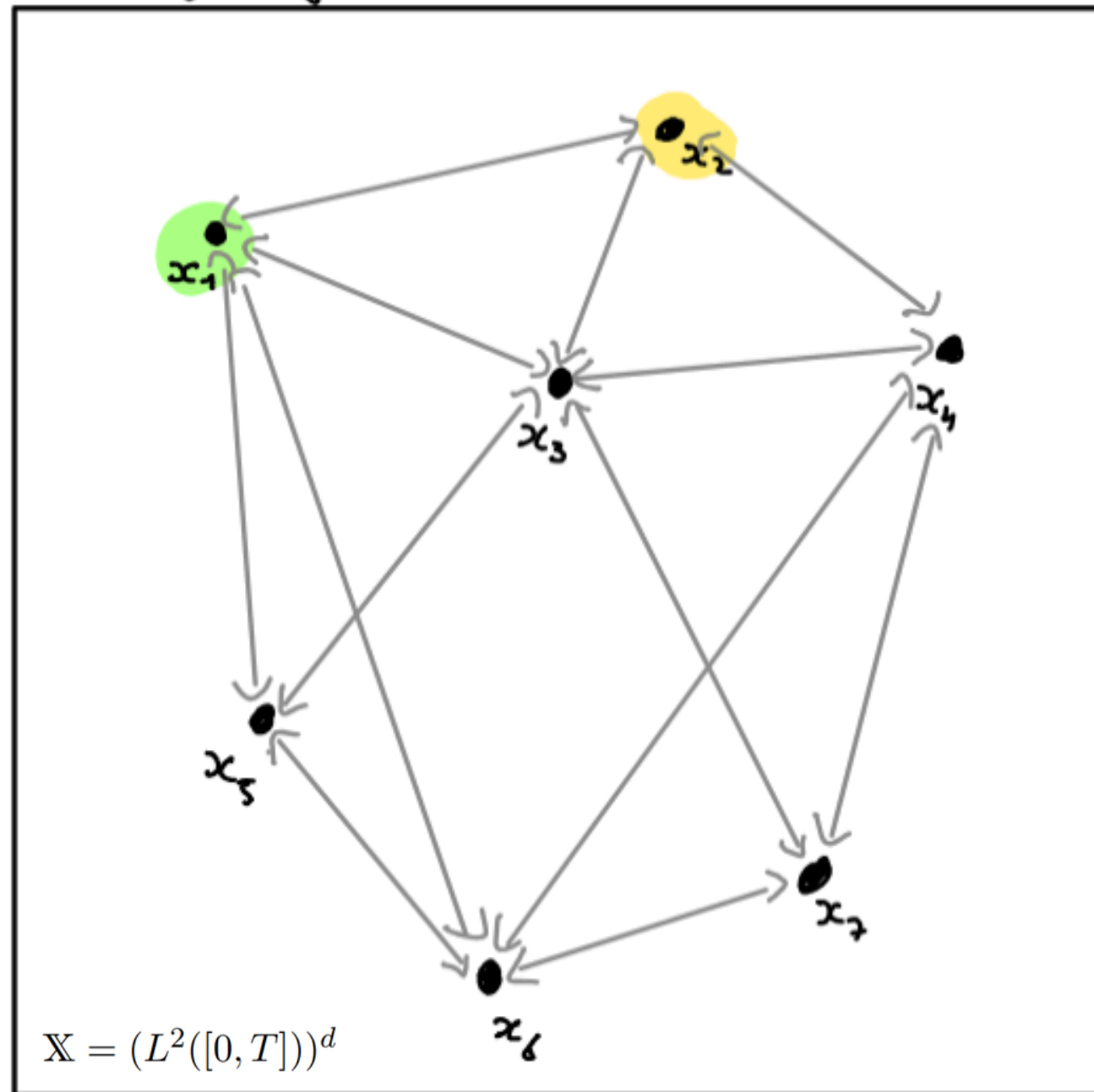
radial kernel



$$k(X, X') = \sigma^2 \exp\left(-\frac{1}{2\ell^2} \|X - X'\|_{H^1}^2\right)$$

Adapt Design of Experiment

Design of Experiment



$$\text{dist}(x_1, x_2) = \sqrt{\sum_{i=1}^8 \int (f_i(u) - g_i(u))^2 du}$$

$$x_1 = (f_1, f_2, \dots, f_8)$$

$$x_2 = (g_1, g_2, \dots, g_8)$$

$\vdots$

Maxi-min DoE :
maximize the
minimum distance

Randomly generate
1000 designs of 200 points each

Our adaptation still gives

accurate prediction

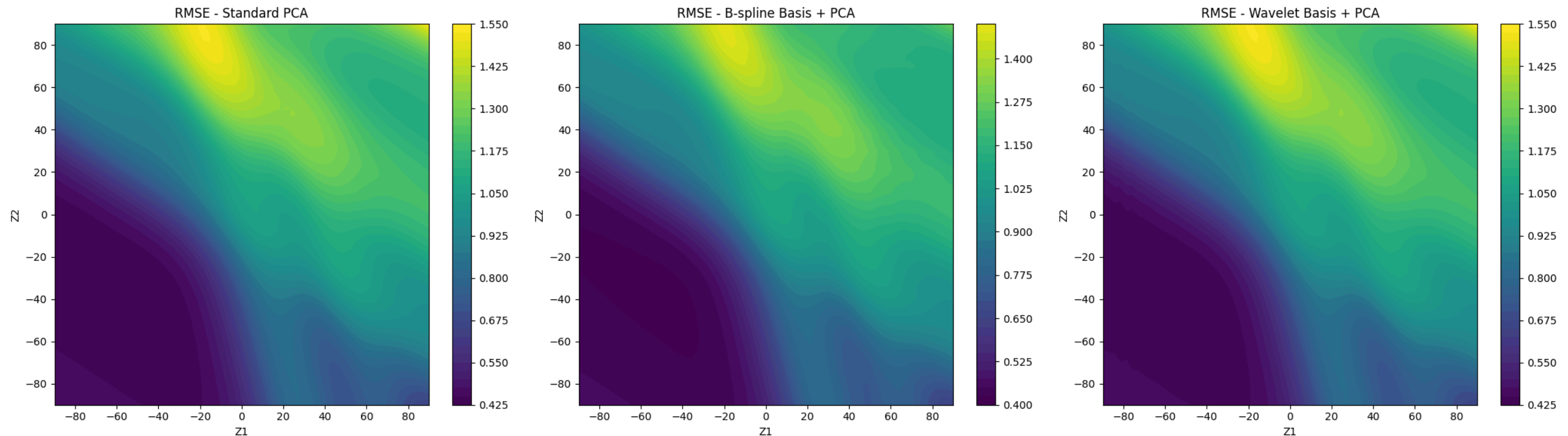
- Generate train and test data ✓
- Find a nice DoE ✓
- Use gaussian processes ✓
- Reduce output dimension ✓
(nothing to change)

I

Methodology and toy model

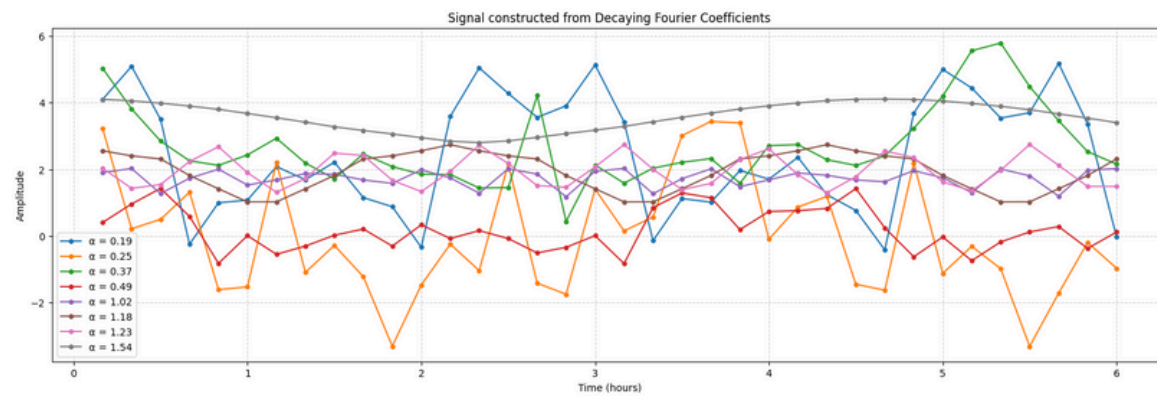
III

RMSE maps for minimum
water level

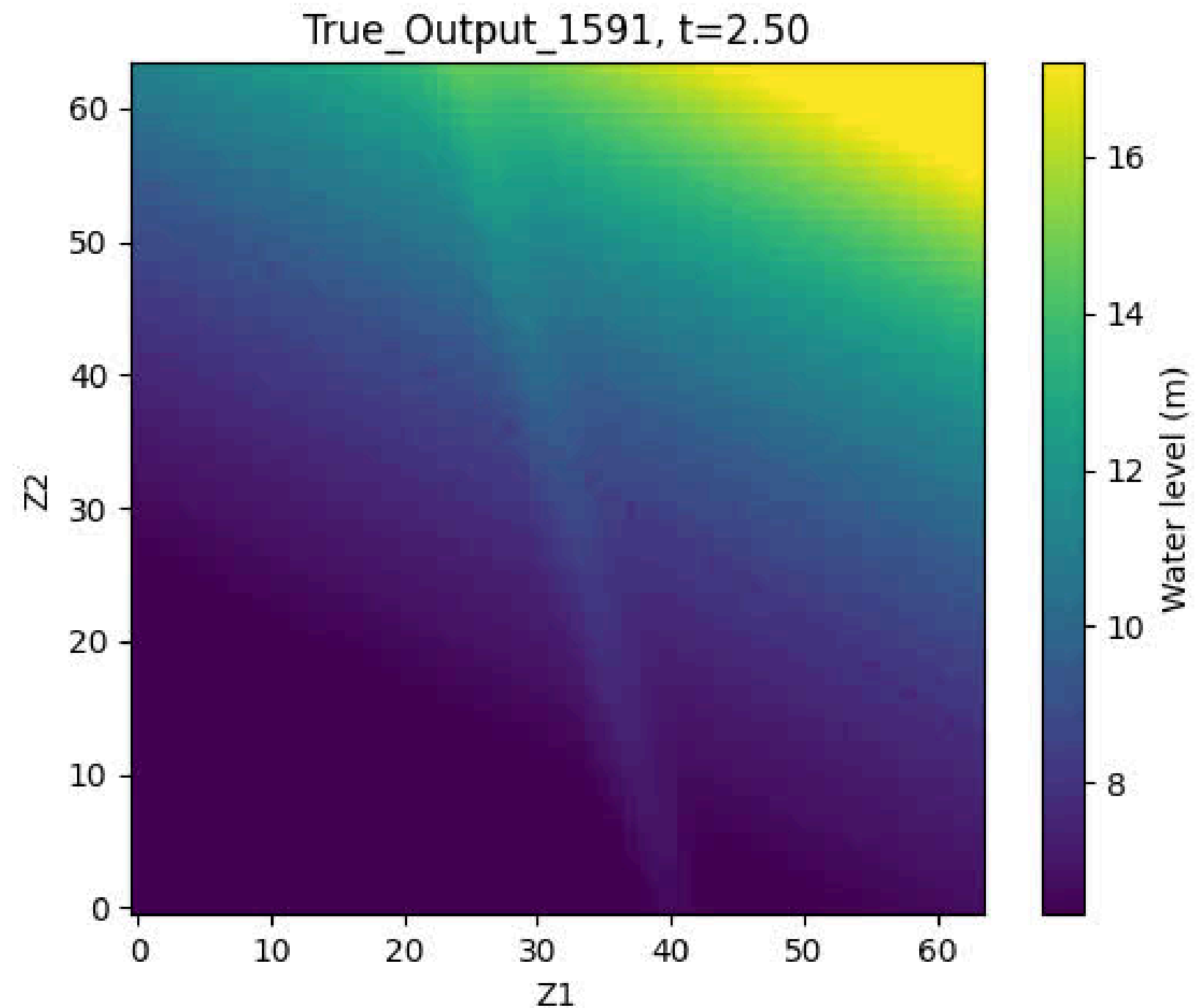


Adding temporality to the output

$$X = (f_1, f_2, \dots, f_8) \quad \mathbb{X} = (L^2([0, T]))^d$$



Compute Campbell
at each time step



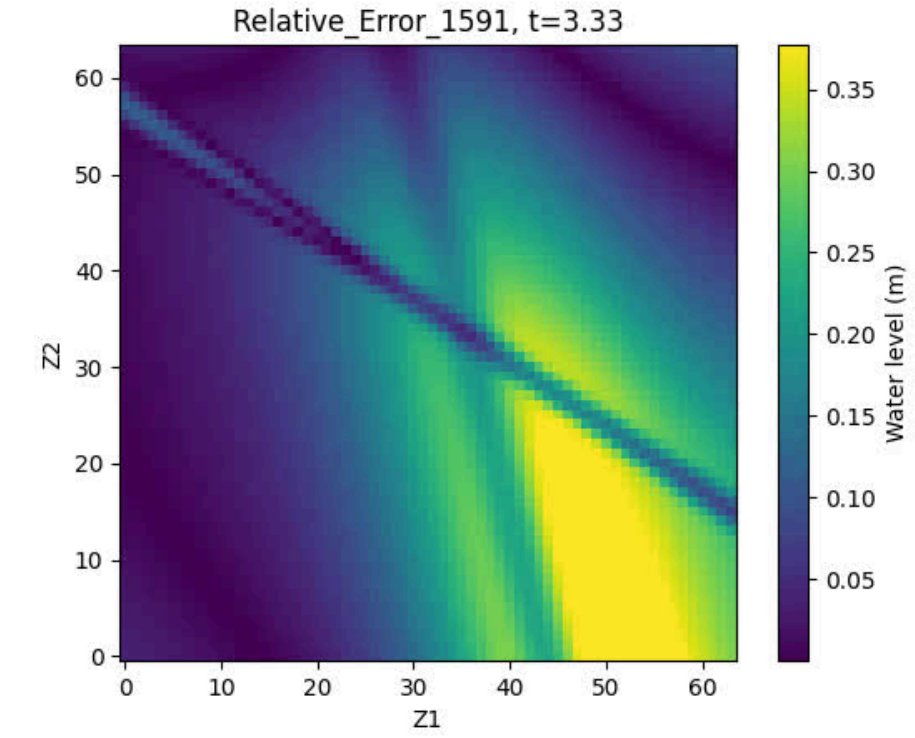
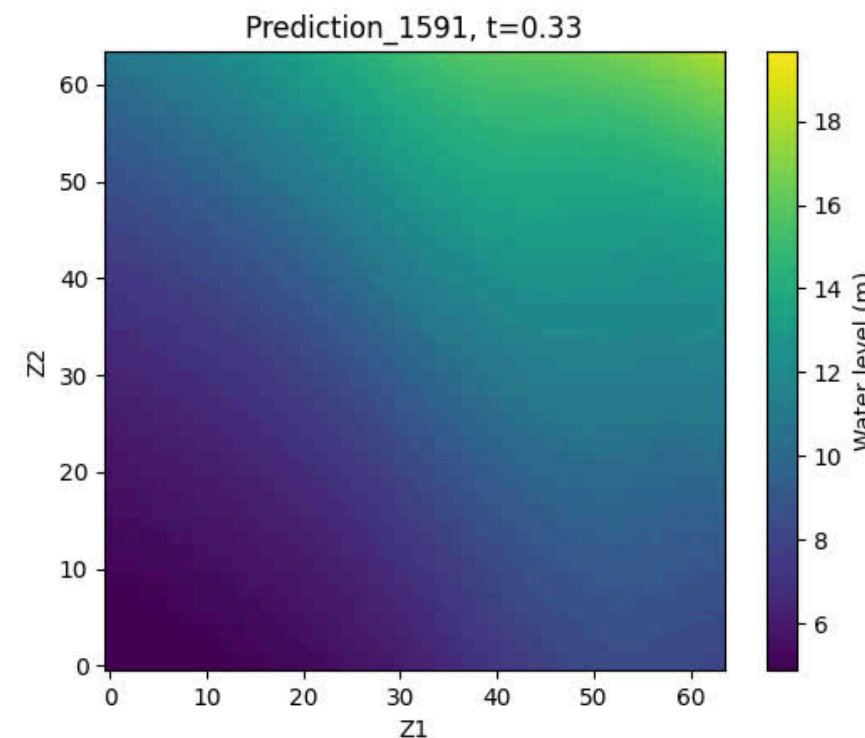
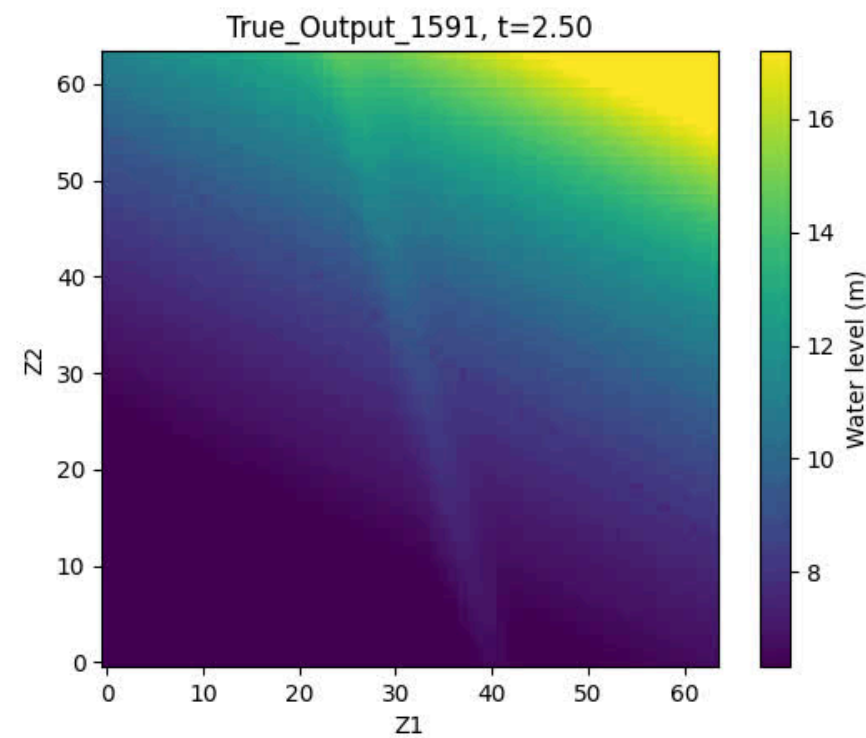
Difficulty to be accurate when **time discontinuity**

I

Methodology and toy model

III

- Generate train and test data ✓
- Find a nice DoE ✓
- Use gaussian processes ✓
- Reduce output dimension ✓

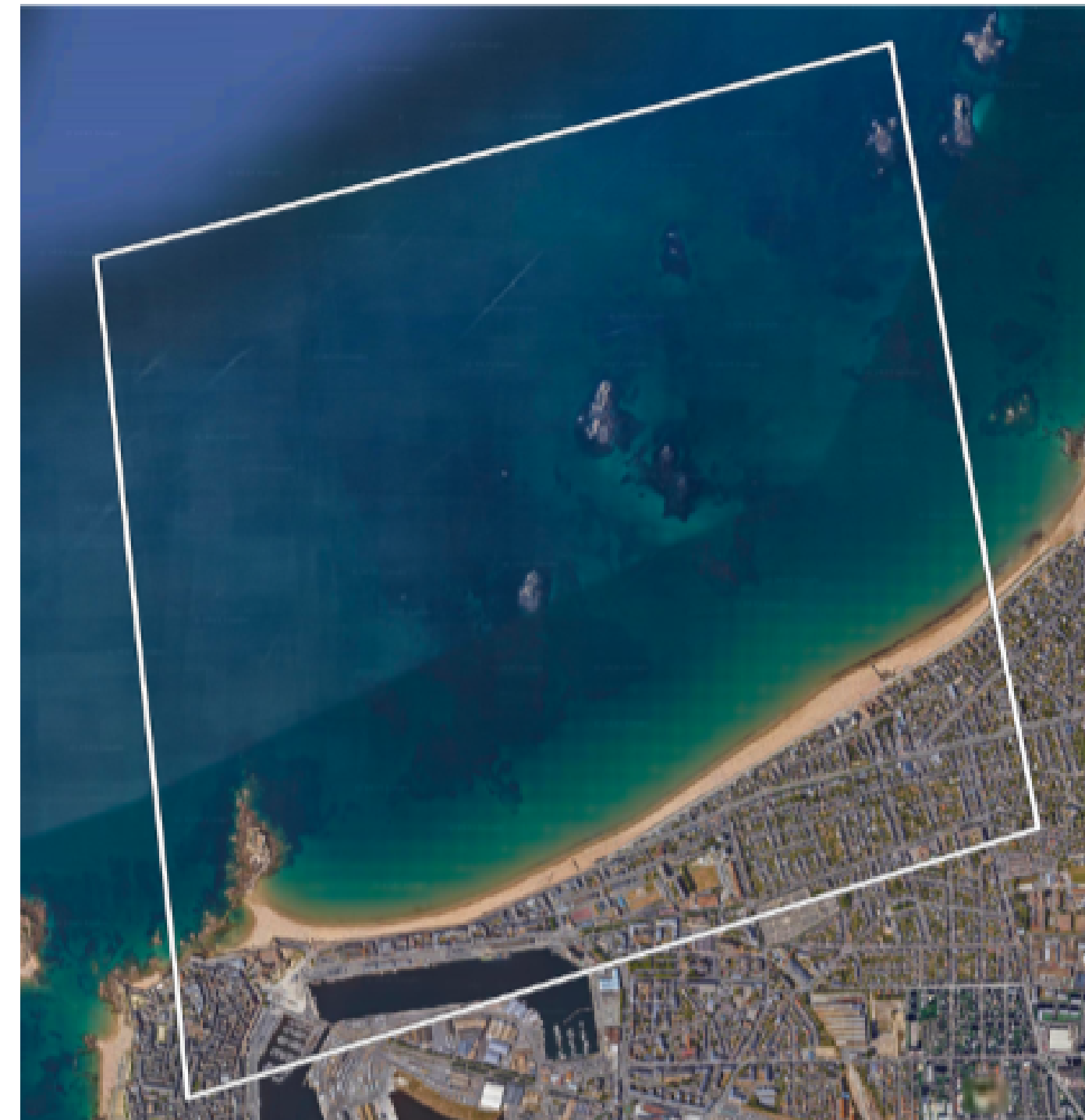


Smooth prediction can't adapt to time discontinuity/high variations

Saint Malo

- Context
- Application to scalar scenario
- Application on functional input/output

Study zone



Study area of the Bay of Saint-Malo, as seen from above

Compute simulations using **Tolosa**

I

II

Saint-Malo

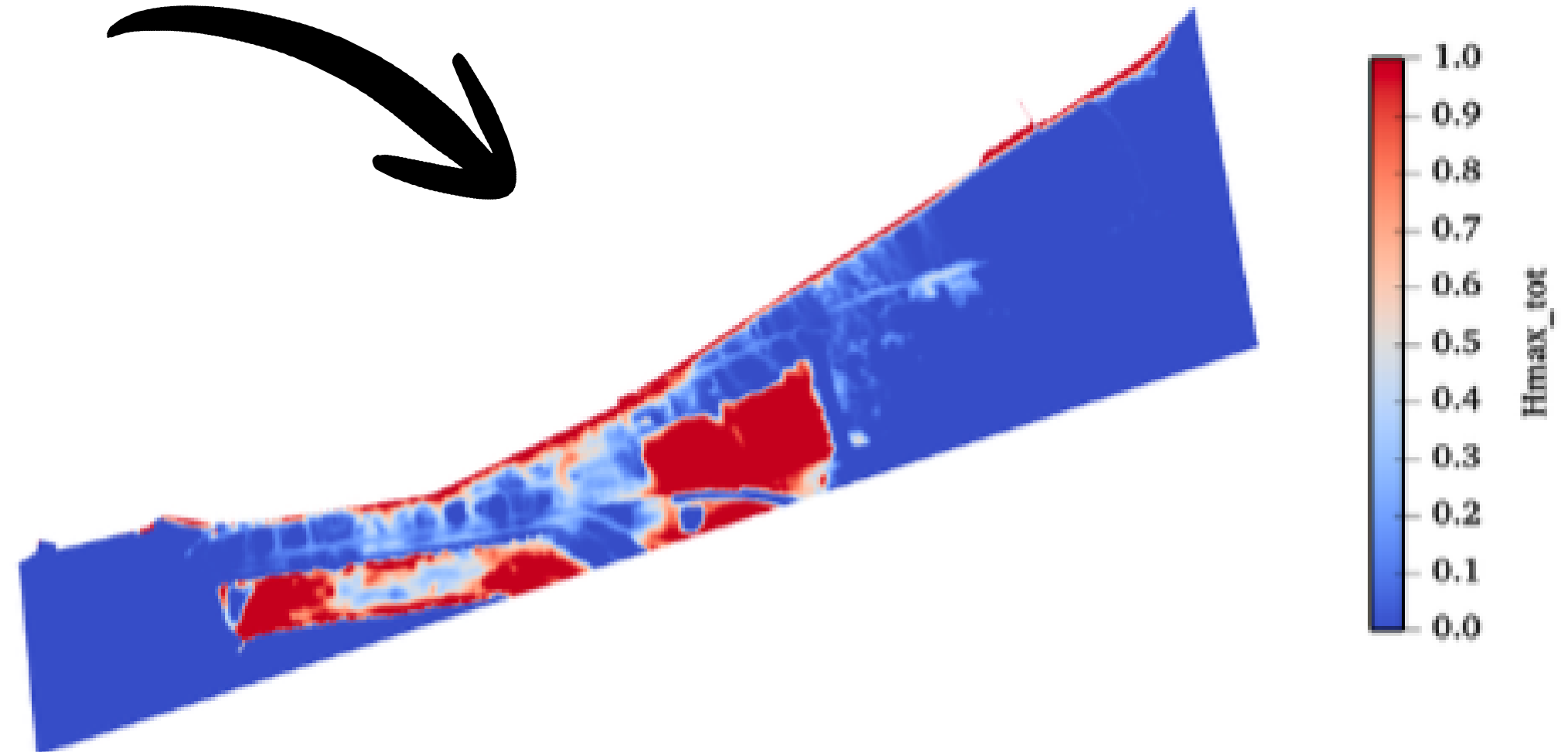
Input Data :

- Mean Sea Level (MSL) from 0 to 1 meter
- Significant Wave Height (H_s) from 1 to 3 meters



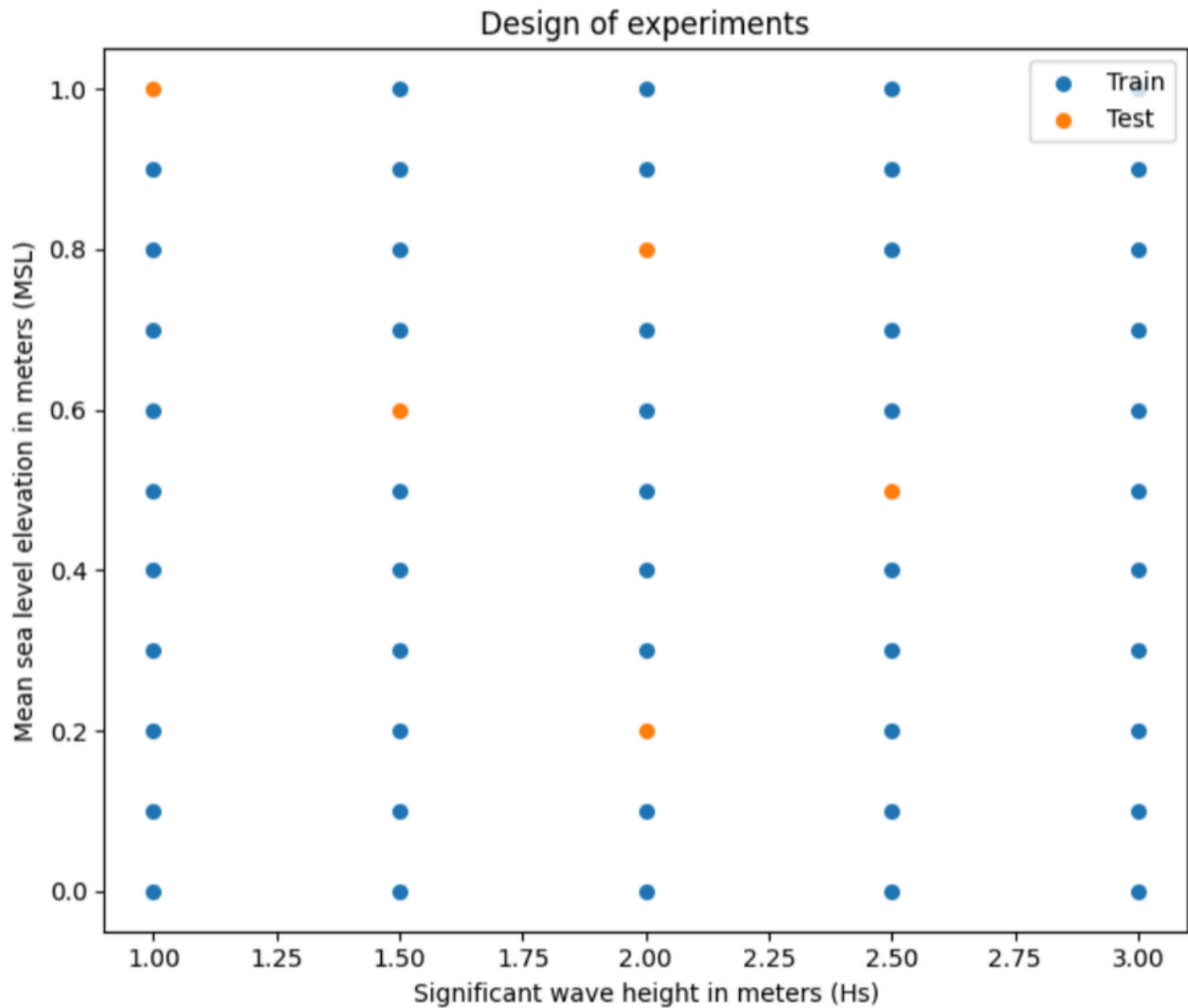
High computation
time cost !!!

**Tolosa
simulator**



Visualization of output given by the Tolosa simulator

Accurate prediction with suboptimal DoE



- Uniform grid with 55 points (5 values of Hs, 11 values of MSL)
- 50 simulations for training and 5 to test
- Imposed DoE
- 14 hours to compute simulations on this DoE

Wavelet metamodel with 5 principal components



Dataset	RMSE
Train	0.054
Test	0.084

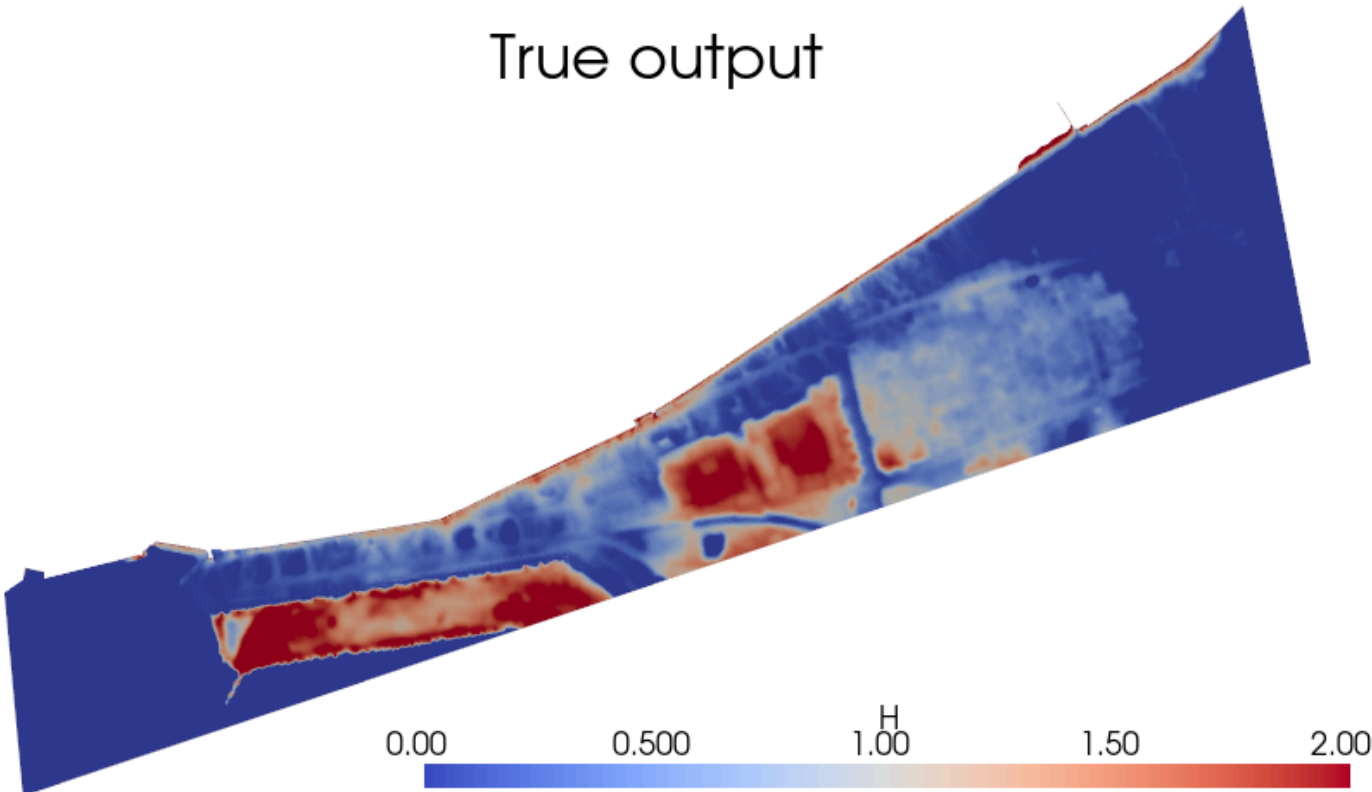
Accurate where water is high, less where water is low

I

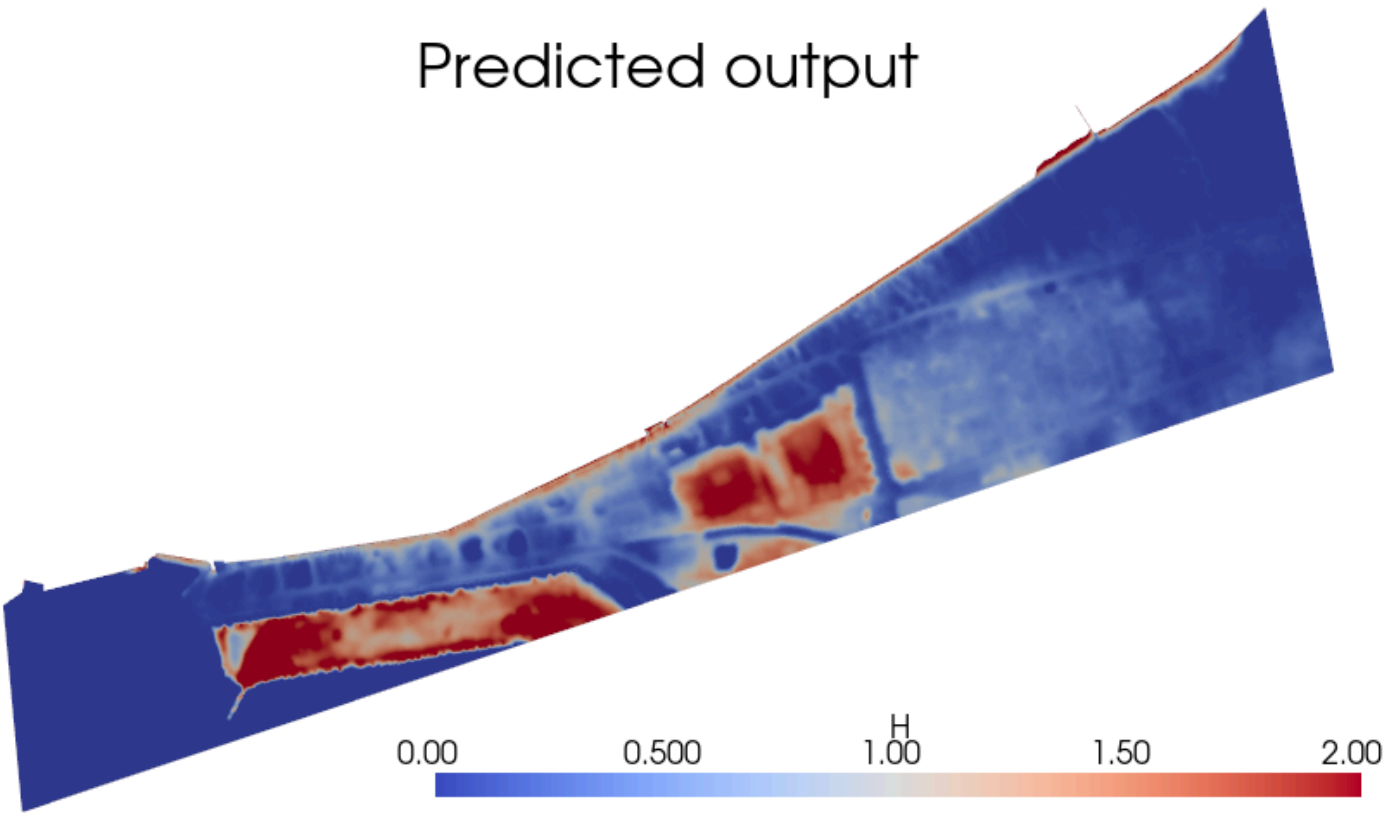
II

Saint-Malo

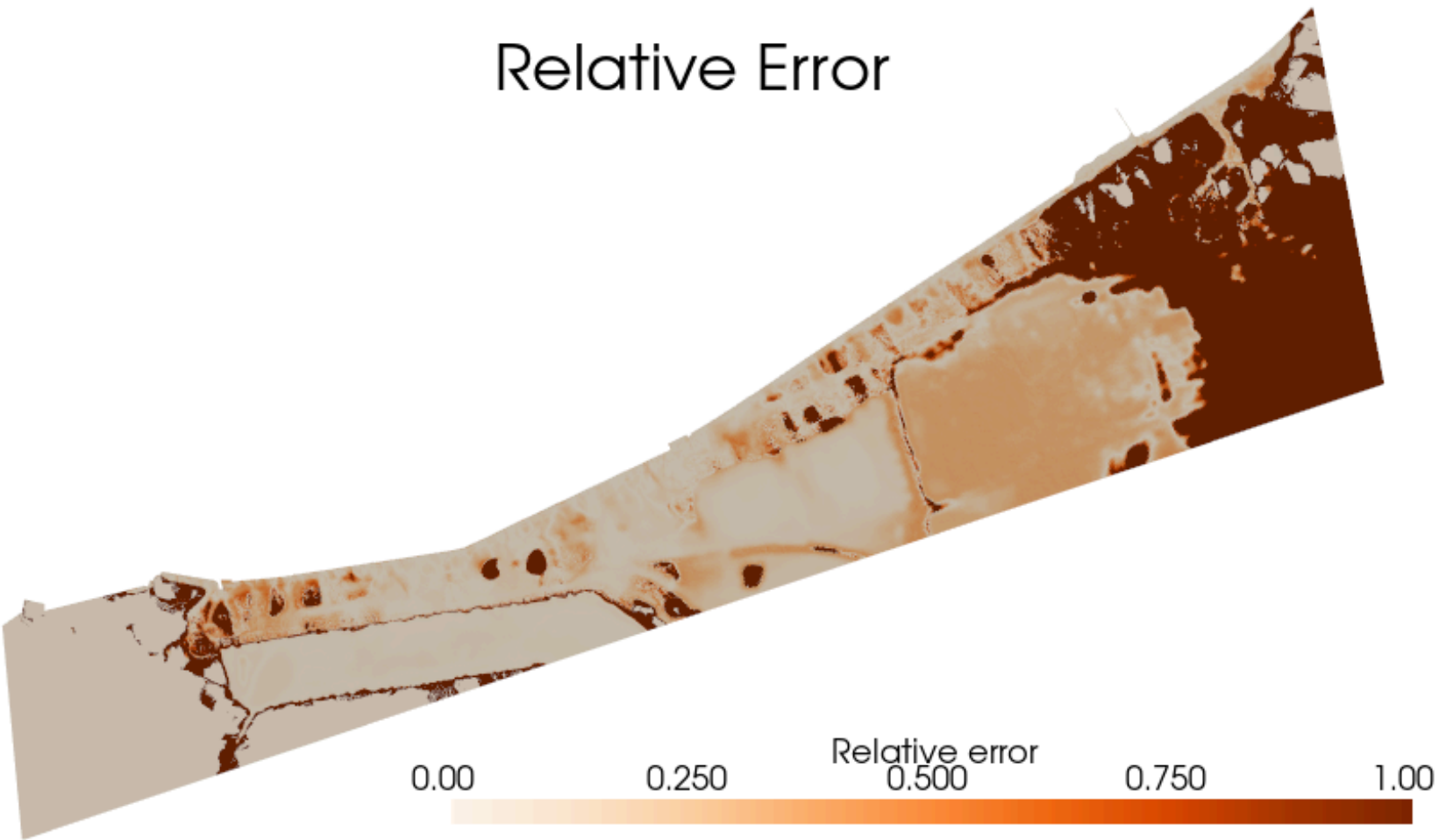
True output



Predicted output



Relative Error

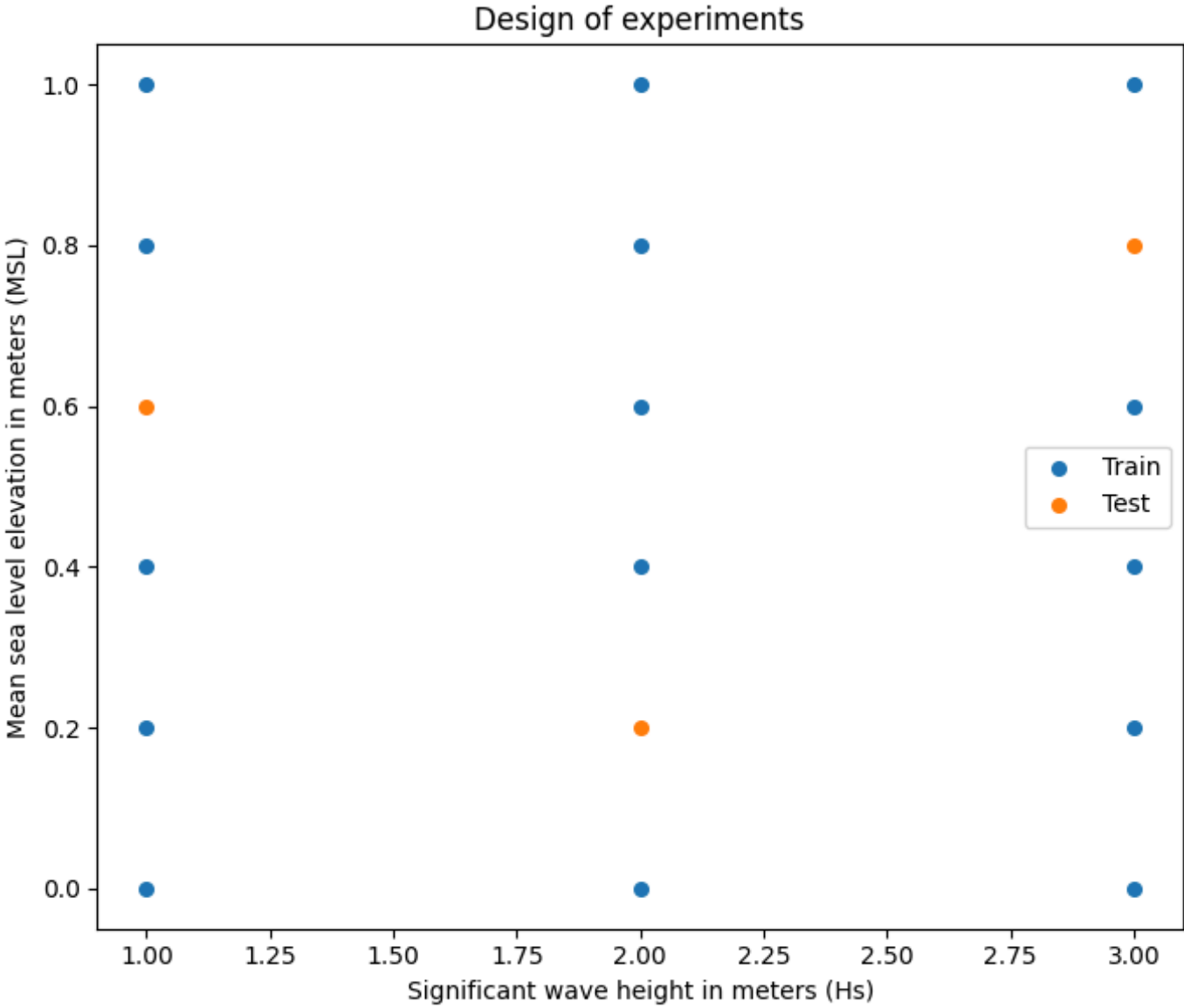


Adding temporality, but less data

I

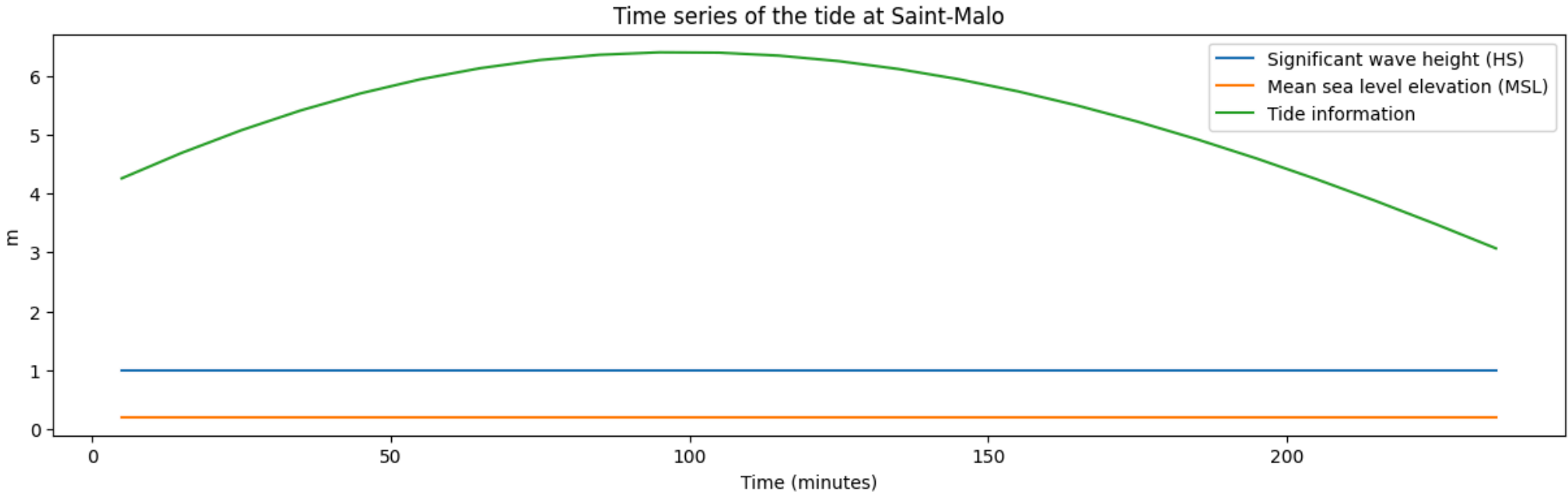
II

Saint-Malo



- Uniform grid with 18 points (5 values of Hs, 11 values of MSL)
- 15 simulations for training and 3 to test
- Imposed DoE
- 5 hours to compute simulations on this DoE

Temporality



- Hs and MSL : constant function of time
- Tide : varying through time

Accurate where water is high, less where water is low

I

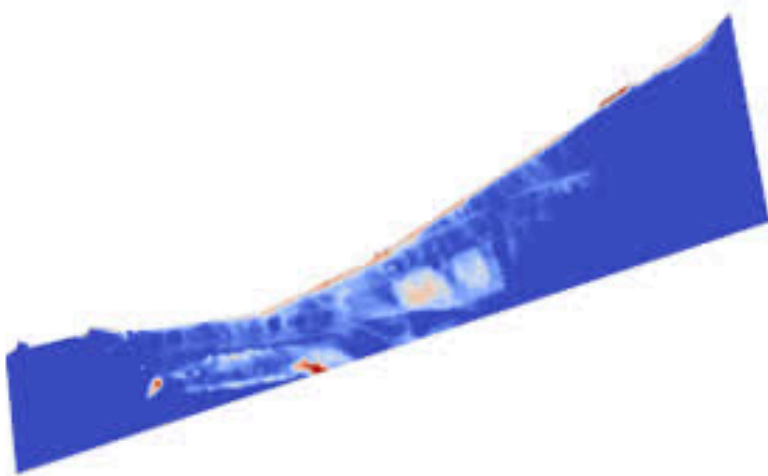
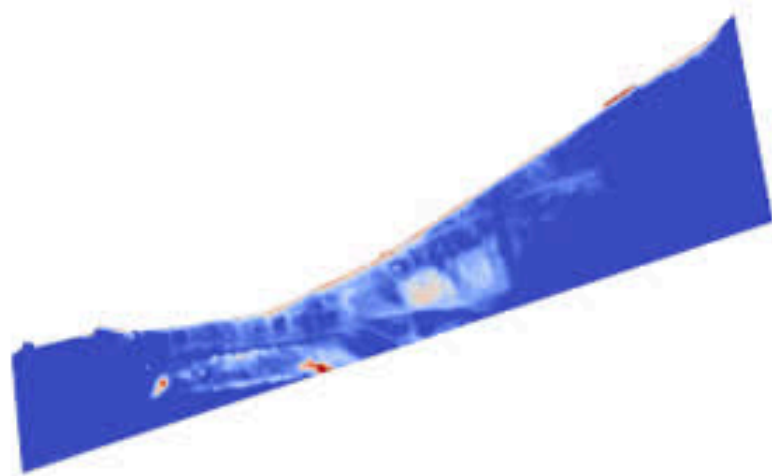
II

Saint-Malo

True Output (t=5/24)

Prediction (t=5/24)

Error (t=5/24)



Conclusion

Real-time prediction thanks to Gaussian Processes metamodel

Encouraging results on both toy model and real-case study

To look into:

- exploiting variance
- DoE
- spatio-temporal dimension reduction

Contacts

- Paul Slisse , slisse@insa-toulouse.fr



- Victor Davodeau , davodeau@insa-toulouse.fr



<https://github.com/Viictordvd/Vigilance-Vagues-Submersion>

Accurate where water is high, less were water is low

I

II

Saint-Malo

